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Sugiura; Yoshinori et al.

Fiber-reinforced resin molded body and production method thereof, fiber-reinforced resin molding prepreg, fiber-reinforced molded body and production method of fiber-reinforced molded body and resin sheet, fiber-reinforced sandwich composite, and production method of fiberreinforced molded body

Abstract

In a method for producing a fiber-reinforced resin molded body (10) by heat-compressing fiber substrates (11A to 11D) together with a thermosetting resin (15) so that the thermosetting resin (15) is impregnated into the fiber substrates (11A to 11D) and cured, a thermosetting resin powder (15A) is disposed in contact with at least one surface of the fiber substrates (11A to 11D), the fiber substrates (11A to 11D) are heat-compressed together with the thermosetting resin powder (15A) by a mold (30) so that the thermosetting resin powder (15A) is melted, impregnated into the fiber substrates (11A to 11D), and cured. Also disclosed is a fiber-reinforced resin molded body as well as a vehicle or airframe including a fiber-reinforced resin molded body.

Inventors:	Sugiura; Yoshinori (Anjo, JP), Harata; Naoya (Anjo, JP), Yasui; Tatsuhiko (Anjo, JP), Tanabe; Naoyuki (Kaizu, JP)
Applicant:	INOAC CORPORATION (Nagoya, JP)
Family ID:	1000008819808
Assignee:	INOAC CORPORATION (Nagoya, JP)
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Primary Examiner: Khatri; Prashant J

Attorney, Agent or Firm: Sughrue Mion, PLLC

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

(1) This application is a National Stage of International Application No. PCT/JP2021/023675 filed Jun. 22, 2021, claiming priority based on Japanese Patent Application No. 2020-116251 filed Jul. 6, 2020; Japanese Patent Application No. 2020-171081 filed Oct. 9, 2020; Japanese Patent Application No. 2020-171814 filed Oct. 12, 2020, Japanese Patent Application No. 2020-206600 filed Dec. 14, 2020, and Japanese Patent Application No. 2021-076007 filed Apr. 28, 2021, the entire contents of which are incorporated herein by reference.

TECHNICAL FIELD

(2) The present disclosure relates to: a fiber-reinforced resin molded body and a method for producing the fiber-reinforced resin molded body; a fiber-reinforced resin molding prepreg; a fiber-reinforced molded body, a method for producing the fiber-reinforced molded body, and a resin sheet; a fiber-reinforced sandwich composite; and a method for producing the fiber-reinforced molded body.

BACKGROUND ART

(3) In recent years, for the purpose of weight reduction and improvement of mechanical strength, fiber-reinforced resin molded bodies made of a composite material including: a fiber substrate made of carbon fiber, glass fiber or the like; and a thermosetting resin are widely used in various fields and applications.

(4) In particular, in transportation equipment such as automobiles, railways, and airplanes, there is a high demand for low fuel consumption, and the effect of low fuel consumption provided by weight reduction of vehicles and airframes is high. Therefore, fiber-reinforced resin molded bodies having excellent lightweight properties are expected as alternative materials for metal.

(5) As a method for producing a fiber-reinforced resin molded body, there is a method in which a fiber substrate is impregnated with a thermosetting resin to form a prepreg, and then the prepreg is molded using an autoclave, a hot press, or the like.

(6) The thermosetting resin which is impregnated into the fiber substrate in the preparation of the prepreg is generally liquid. However, a liquid resin has a problem of a pot life, and there is also a problem of working environment and air pollution when a solvent is used.

(7) As a method for solving these problems, a prepreg prepared by using a powder resin has been proposed (Patent Literature 1).

(8) In addition, as a molding method not using a prepreg, there is a method (RTM method) in which a preform is prepared from a fiber substrate, the preform is inserted into a mold, and then a liquid resin is injected into the mold and cured to obtain a molded body.

(9) In addition, a lightweight fiber-reinforced sandwich composite obtained by laminating a prepreg and a core component and heat-pressing the laminate has been proposed (Patent Literature

2).

CITATIONS LIST

Patent Literature

(10) Patent Literature 1: JP 2006-232915 A Patent Literature 2: JP 2020-44811 A

SUMMARY OF INVENTION

Technical Problems

(11) However, in the molding method using a prepreg, regardless of whether a liquid resin or a powdery resin is used, there is a problem that the production cost of the fiber-reinforced resin molded body increases because a large-scale facility is required for a process of forming the prepreg, and management of the process of forming the prepreg is complicated.

(12) Further, the RTM method not using a prepreg requires preparation of a preform, and requires equipment for injecting a liquid resin into a mold at a high pressure in a predetermined amount. Thus, the RTM method has a problem that the production cost of the fiber-reinforced resin molded body increases as in the case of using the prepreg.

(13) In addition, a prepreg formed by using a conventional powder resin (Patent Literature 1) requires a complicated process of once melt-kneading a plurality of solid epoxy resins, a tetracarboxylic acid dianhydride, and a curing accelerator to prepare an epoxy resin composition, and then pulverizing the epoxy resin composition to prepare a powder resin, which is a factor of cost increase. In addition, since the epoxy resin composition as a base of the powder resin contains the curing accelerator, there is a possibility that curing by the curing accelerator may proceed even after preparation of the prepreg formed by using the powder resin. Therefore, such a prepreg does not have sufficient storage stability, and may affect the quality of a fiber-reinforced resin molded body molded from the prepreg.

(14) In addition, the conventional prepreg has poor storage stability because a curing reaction gradually proceeds even at normal temperature during storage until use. Such a prepreg with poor storage stability may affect the quality of a fiber-reinforced resin molded body or a sandwich structure obtained using the stored prepreg.

(15) The present disclosure has been made to solve at least one of the above-described problems.

Solutions to Problems

First Aspect

(16) A fiber-reinforced resin molded body obtained by heat-compressing a fiber substrate together with a thermosetting resin so that the thermosetting resin is impregnated into the fiber substrate and cured, wherein the thermosetting resin cured is obtained by melting a powder of the thermosetting resin disposed in contact with the fiber substrate and curing the thermosetting resin during the heat compression.

(17) The fiber-reinforced molded body wherein the fiber substrate is composed of a plurality of layers.

(18) A method for producing a fiber-reinforced resin molded body by heat-compressing a fiber substrate together with a thermosetting resin so that the thermosetting resin is impregnated into the fiber substrate and cured, the method including: disposing a powder of the thermosetting resin in contact with the fiber substrate; and heat-compressing the fiber substrate together with the powder of the thermosetting resin by a mold so that the powder of the thermosetting resin is melted, impregnated into the fiber substrate, and cured.

(19) The method for producing a fiber-reinforced resin molded body, wherein the fiber substrate has a plurality of layers, and the powder of the thermosetting resin is disposed on at least one surface of the plurality of layers of the fiber substrate.

(20) The method for producing a fiber-reinforced resin molded body, wherein a temperature T_c ° C. of the mold is a temperature satisfying

$[T_b + (T_b - T_a)/3] - 15 \leq T_c \leq [T_b + (T_b - T_a)/3] + 20$, wherein a melting start temperature of the thermosetting resin is T_a ° C. and a curing reaction start temperature of the thermosetting resin is

Tb° C.

(21) The method for producing a fiber-reinforced resin molded body, wherein a value of (Tb-Ta) of the thermosetting resin satisfies

$30 \leq (Tb - Ta) \leq 100$, wherein a melting start temperature of the thermosetting resin is Ta° C. and a curing reaction start temperature of the thermosetting resin is Tb° C.

(22) The method for producing a fiber-reinforced resin molded body, wherein the thermosetting resin has a minimum viscosity of 2,000 Pa.Math.s or less at a melting start temperature of Ta° C. or higher.

(23) The method for producing a fiber-reinforced resin molded body, wherein the thermosetting resin has a maximum viscosity of 1,000 Pa.Math.s or more in a temperature range of a curing reaction start temperature Tb° C. to 190° C.

(24) The method for producing a fiber-reinforced resin molded body, wherein the thermosetting resin has a melting start temperature Ta° C. of 60 to 100° C.

(25) The method for producing a fiber-reinforced resin molded body, wherein the thermosetting resin is a resin selected from the group consisting of a phenol resin, a mixed resin of a phenol resin and an epoxy resin, a mixed resin of a phenol resin and a cyanate resin, and a mixed resin of a phenol resin, a cyanate resin, and an epoxy resin.

Second Aspect

(26) A fiber-reinforced resin molding prepreg, the prepreg being obtained by heat-compressing a fiber substrate together with a thermosetting resin, wherein the thermosetting resin before heat compression is in a form of a powder, and the thermosetting resin has a viscosity of 2,000 Pa.Math.s or less at a curing reaction start temperature Tb° C., and a maximum viscosity of 1,000 Pa.Math.s or more in a temperature range of the curing reaction start temperature Tb° C. to 190° C.

(27) The fiber-reinforced resin molding prepreg, wherein the thermosetting resin has a melting start temperature Ta° C. of 60 to 100° C.

(28) The fiber-reinforced resin molding prepreg, wherein in the thermosetting resin, a value obtained by subtracting the melting start temperature Ta° C. from the curing reaction start temperature Tb° C. satisfies

$30 \leq (Tb - Ta) \leq 100$.

(29) The fiber-reinforced resin molding prepreg, wherein a melting start temperature Ta° C. of the thermosetting resin is equal to or lower than a temperature during the heat compression, and the curing reaction start temperature Tb° C. of the thermosetting resin is equal to or higher than the temperature during the heat compression.

(30) The fiber-reinforced resin molding prepreg, wherein in the thermosetting resin, the melting start temperature Ta° C. is the temperature during the heat compression minus 5° C. or lower, and the curing reaction start temperature Tb° C. is the temperature during heat compression plus 5° C. or higher.

(31) The fiber-reinforced resin molding prepreg, wherein the thermosetting resin is a resin selected from the group consisting of a phenol resin, a mixed resin of a phenol resin and an epoxy resin, a mixed resin of a phenol resin and a cyanate resin, and a mixed resin of a phenol resin, an epoxy resin, and a cyanate resin.

(32) A fiber-reinforced resin molded body obtained by curing the fiber-reinforced resin molding prepreg through heat compression.

Third Aspect

(33) A fiber-reinforced molded body including: a fiber substrate; and a resin sheet containing a thermosetting resin, the fiber substrate being integrated by the thermosetting resin of the resin sheet, wherein the thermosetting resin has a viscosity of 2,000 Pa.Math.s or less at a curing reaction start temperature Tb° C., and a maximum viscosity of 1,000 Pa.Math.s or more in a temperature range of the curing reaction start temperature Tb° C. to 190° C.

(34) The fiber-reinforced molded body, wherein the resin sheet includes a sheet substrate.

(35) The fiber-reinforced molded body, wherein, in the thermosetting resin, a value of $(T_b - T_a)$ satisfies

$30 \leq (T_b - T_a) \leq 100$, wherein a melting start temperature of the thermosetting resin is $T_a^\circ \text{C.}$ and a curing reaction start temperature of the thermosetting resin is $T_b^\circ \text{C.}$

(36) The fiber-reinforced molded body, wherein the thermosetting resin is a resin selected from the group consisting of a phenol resin, a mixed resin of a phenol resin and an epoxy resin, a mixed resin of a phenol resin and a cyanate resin, a mixed resin of an epoxy resin and a cyanate resin, and a mixed resin of a phenol resin, an epoxy resin, and a cyanate resin.

(37) A method for producing the fiber-reinforced molded body, the method including: heat-compressing the fiber substrate and the resin sheet by a mold in a state in which the fiber substrate and the resin sheet are stacked, so that the thermosetting resin is impregnated into the fiber substrate and cured.

(38) The method for producing a fiber-reinforced molded body, wherein a temperature $T_c^\circ \text{C.}$ during heat compression is a temperature satisfying

$[T_b + (T_b - T_a)/3] - 15 \leq T_c \leq [T_b + (T_b - T_a)/3] + 20$.

(39) A resin sheet for producing a fiber-reinforced molded body, the resin sheet containing a thermosetting resin.

Fourth Aspect

(40) A fiber-reinforced sandwich composite including: a fiber substrate; and a core component, the fiber substrate and the core component being bonded and integrated by being laminated and heat-compressed together with a thermosetting resin, wherein the core component has a 5% compressive stress of 0.10 MPa or more, the thermosetting resin before heating is in a form of a powder, and the thermosetting resin has a viscosity of 2,000 Pa.Math.s or less at a curing reaction start temperature $T_b^\circ \text{C.}$, and a maximum viscosity of 1,000 Pa.Math.s or more in a temperature range of the curing reaction start temperature $T_b^\circ \text{C.}$ to 190°C.

(41) The fiber-reinforced sandwich composite, wherein a melting start temperature $T_a^\circ \text{C.}$ and a curing reaction start temperature $T_b^\circ \text{C.}$ of the thermosetting resin are temperatures satisfying $[T_b + (T_b - T_a)/3] - 15 \leq T_c \leq [T_b + (T_b - T_a)/3] + 20$ with respect to a temperature $T_c^\circ \text{C.}$ during heat compression.

(42) The fiber-reinforced sandwich composite, wherein in the thermosetting resin, a value obtained by subtracting the melting start temperature $T_a^\circ \text{C.}$ from the curing reaction start temperature $T_b^\circ \text{C.}$ satisfies

$30 \leq (T_b - T_a) \leq 100$.

(43) The fiber-reinforced sandwich composite, wherein the thermosetting resin has a melting start temperature $T_a^\circ \text{C.}$ of 60 to 100°C.

(44) The fiber-reinforced sandwich composite, wherein the core component is a foam having a closed cell structure.

(45) The fiber-reinforced sandwich composite, wherein the thermosetting resin is a resin selected from the group consisting of a phenol resin, a mixed resin of a phenol resin and an epoxy resin, a mixed resin of a phenol resin and a cyanate resin, and a mixed resin of a phenol resin, a cyanate resin, and an epoxy resin.

Fifth Aspect

(46) A method for producing a fiber-reinforced molded body including a core material obtained by impregnating a thermosetting resin into a resin foam and curing the thermosetting resin; and a fiber reinforcing material obtained by impregnating the thermosetting resin into a fiber substrate and curing the thermosetting resin, the core material and the fiber reinforcing material being laminated and integrated, wherein the resin foam has an open-cell structure, the method including: forming a pre-molding laminate with the resin foam having an open-cell structure, the thermosetting resin, and the fiber substrate; and heat-compressing the pre-molding laminate so that the thermosetting resin is melted, and the thermosetting resin melted is impregnated into the resin foam having an

open-cell structure and the fiber substrate, and cured.

(47) The method for producing a fiber-reinforced molded body, wherein in the pre-molding laminate, the fiber substrate is disposed on both sides of the resin foam having an open-cell structure.

(48) The method for producing a fiber-reinforced molded body, wherein the thermosetting resin in the pre-molding laminate is in a form of a powder, and has a viscosity of 2,000 Pa.Math.s or less at a curing reaction start temperature $T_b^{\circ}\text{C.}$ and a maximum viscosity of 1,000 Pa.Math.s or more in a temperature range of the curing reaction start temperature $T_b^{\circ}\text{C.}$ to 190°C.

(49) The method for producing a fiber-reinforced molded body, wherein a melting start temperature $T_a^{\circ}\text{C.}$ and a curing reaction start temperature $T_b^{\circ}\text{C.}$ of the thermosetting resin are temperatures satisfying

$[T_b + (T_b - T_a)/3] - 15 \leq T_c \leq [T_b + (T_b - T_a)/3] + 20$ in a relationship between a temperature $T_c^{\circ}\text{C.}$ during heat compression.

(50) The method for producing a fiber-reinforced molded body, wherein in the thermosetting resin, a value obtained by subtracting the melting start temperature $T_a^{\circ}\text{C.}$ from the curing reaction start temperature $T_b^{\circ}\text{C.}$ satisfies

$30 \leq (T_b - T_a) \leq 100.$

(51) The method for producing a fiber-reinforced molded body, wherein the thermosetting resin has a melting start temperature $T_a^{\circ}\text{C.}$ of 60 to 100°C.

(52) The method for producing a fiber-reinforced molded body, wherein a resin proportion defined by Equation (A1) is 50 to 80%.

[MathematicalFormula1]

$$(53) \quad \text{Resin proportion} = \frac{\text{Weight of fiber-reinforced molded body} - \text{Total weight of fibers substrate and resin foam before molding}}{\text{Weight of fiber-reinforced molded body}} \times 100 \quad (A1)$$

(54) The method for producing a fiber-reinforced molded body, wherein a compression percentage defined by Equation (A2) is 200 to 5,000%.

[MathematicalFormula2]

$$(55) \quad \text{Compression percentage} = \frac{\text{Thickness of resin foam before molding} - \text{Thickness of core material of fiber-reinforced molded body}}{\text{Thickness of core material of fiber-reinforced molded body}} \times 100 \quad (A2)$$

(56) The method for producing a fiber-reinforced molded body, wherein the thermosetting resin is a resin selected from the group consisting of a phenol resin, a mixed resin of a phenol resin and an epoxy resin, a mixed resin of a phenol resin and a cyanate resin, and a mixed resin of a phenol resin, a cyanate resin, and an epoxy resin.

(57) A fiber-reinforced molded body including: a core material obtained by impregnating a thermosetting resin into a resin foam having an open-cell structure and curing the thermosetting resin; and a fiber reinforcing material obtained by impregnating a fiber substrate with the thermosetting resin and curing the thermosetting resin, the core material and the fiber reinforcing material being laminated and integrated, wherein the thermosetting resin has a viscosity of 2,000 Pa.Math.s or less at a curing reaction start temperature $T_b^{\circ}\text{C.}$, and a maximum viscosity of 1,000 Pa.Math.s or more in a temperature range of the curing reaction start temperature $T_b^{\circ}\text{C.}$ to 190°C.

Advantageous Effects of Invention

First Aspect

(58) According to the fiber-reinforced resin molded body of the first aspect, by using the powder of the thermosetting resin disposed in contact with the fiber substrate, a fiber-reinforced resin molded body having high strength and high rigidity can be obtained without requiring a solvent, without a risk of environmental pollution, and without a pot life.

(59) According to the invention of the method for producing a fiber-reinforced resin molded body,

the powder of the thermosetting resin is disposed in contact with the fiber substrate, and the fiber substrate is heat-compressed together with the powder of the thermosetting resin to produce a fiber-reinforced resin molded body. The production method according to the invention thus does not require preparation of prepreg, or preparation of preform as in the RTM method, does not require a solvent for liquid thermosetting resin, which eliminates a risk of environmental pollution, and there is no pot life of the thermosetting resin. As a result, a fiber-reinforced resin molded body having high strength and high rigidity can be produced easily and inexpensively, and safely without the risk of environmental pollution.

(60) When the fiber substrate is heat-compressed together with the powder of the thermosetting resin, with the temperature $T_c^{\circ}\text{C.}$ of the mold set to a temperature satisfying $[T_b + (T_b - T_a)/3] - 15 \leq T_c \leq [T_b + (T_b - T_a)/3] + 20$ with respect to the melting start temperature $T_a^{\circ}\text{C.}$ and the curing reaction start temperature $T_b^{\circ}\text{C.}$ of the thermosetting resin, melting of the powder of the thermosetting resin in contact with the fiber substrate is improved, so that the thermosetting resin is easily impregnated into the fiber substrate. As a result, a fiber-reinforced resin molded body having uniform physical properties can be produced with high production efficiency.

(61) When the thermosetting resin satisfies the relationship of $30 \leq (T_b - T_a) \leq 100$, the melted thermosetting resin can be sufficiently impregnated into the fiber substrate, and a fiber-reinforced resin molded body having uniform physical properties can be produced.

(62) When the thermosetting resin has a minimum viscosity of 2,000 Pa.Math.s or less at the melting start temperature of $T_a^{\circ}\text{C.}$ or higher, the melted thermosetting resin can be sufficiently impregnated into the fiber substrate, and a fiber-reinforced resin molded body having uniform physical properties can be produced.

(63) When the thermosetting resin has a maximum viscosity of 1,000 Pa.Math.s or more in a temperature range of the curing reaction start temperature $T_b^{\circ}\text{C.}$ to 190°C. , the melted thermosetting resin can be impregnated into the fiber substrate and retained therein. As a result, the shapability of the fiber-reinforced resin molded body is good, and sufficient strength can be obtained in a short time.

(64) When the melting start temperature $T_a^{\circ}\text{C.}$ of the thermosetting resin is 60 to 100°C. , temperature control is easy when the fiber substrate is heat-compressed together with the powder of the thermosetting resin to melt and cure the thermosetting resin.

(65) When the thermosetting resin is a resin selected from the group consisting of a phenol resin, a mixed resin of a phenol resin and an epoxy resin, a mixed resin of a phenol resin and a cyanate resin, and a mixed resin of a phenol resin, a cyanate resin, and an epoxy resin, a fiber-reinforced resin molded body excellent in strength and flame retardancy can be obtained.

Second Aspect

(66) According to the means of the second aspect, it is possible to obtain a prepreg that can be prepared simply and inexpensively and has good storage stability, and a fiber-reinforced resin molded body that is prepared from the prepreg and has good quality.

Third Aspect

(67) The fiber-reinforced molded body of the third aspect can be produced at low cost.

Fourth Aspect

(68) According to the fourth aspect, it is possible to obtain a fiber-reinforced sandwich composite that is integrated with a core component and can be produced easily, inexpensively, and safely without a risk of environmental pollution, without using a prepreg and without concern for storage stability of a prepreg.

Fifth Aspect

(69) According to the fifth aspect, a fiber-reinforced molded body can be obtained without using a prepreg. In addition, according to the fifth aspect, a fiber-reinforced molded body shaped into an irregularity shape can also be obtained.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. 1 is a cross-sectional view of a fiber-reinforced resin molded body according to an embodiment of the first aspect.
- (2) FIG. 2 is a cross-sectional view showing lamination and heat compression in an embodiment of a method for producing a fiber-reinforced resin molded body of the first aspect.
- (3) FIG. 3 is a cross-sectional view showing lamination and heat compression in another embodiment of the method for producing a fiber-reinforced resin molded body of the first aspect.
- (4) FIG. 4 is a table showing melting start temperatures, reaction start temperatures, and the like of thermosetting resins used in Examples and Comparative Examples according to the first aspect.
- (5) FIG. 5 is a table showing configurations, physical property values, and the like of Examples and Comparative Examples according to the first aspect.
- (6) FIG. 6 is a graph showing results of viscosity measurement of thermosetting resins used in Examples 1A to 3A and Comparative Examples 1A and 2A according to the first aspect.
- (7) FIG. 7 is a cross-sectional view showing a fiber-reinforced resin molding prepreg according to a first embodiment of the second aspect, and heat compression in the preparation of the prepreg.
- (8) FIG. 8 is a cross-sectional view showing a fiber-reinforced resin molding prepreg according to a second embodiment of the second aspect, and heat compression in the preparation of the prepreg.
- (9) FIG. 9 is a cross-sectional view showing a fiber-reinforced resin molding prepreg according to a third embodiment of the second aspect, and heat compression in the preparation of the prepreg.
- (10) FIG. 10 is a table showing melting start temperatures, curing reaction start temperatures, and the like of thermosetting resins used in fiber-reinforced resin molding prepreps in Examples and Comparative Examples according to the second aspect.
- (11) FIG. 11 is a graph showing results of viscosity measurement of thermosetting resins used in fiber-reinforced resin molding prepreps in Examples 1B to 3B and Comparative Examples 1B and 2B according to the second aspect.
- (12) FIG. 12 is a table showing configurations, physical property values, and the like of fiber-reinforced resin molded bodies prepared from fiber-reinforced resin molding prepreps in Examples and Comparative Examples according to the second aspect.
- (13) FIG. 13 is a table showing appearances, physical properties, and the like of fiber-reinforced resin molded bodies prepared by using fiber-reinforced resin molding prepreps of Examples 1B and 4B to 7B according to the second aspect, the prepreps being used 90 days after preparation.
- (14) FIG. 14 is a cross-sectional view of a fiber-reinforced molded body according to an embodiment of the third aspect.
- (15) FIG. 15 is a cross-sectional view showing lamination and heat compression in an embodiment of a method for producing a fiber-reinforced molded body of the present disclosure.
- (16) FIG. 16 is a cross-sectional view showing lamination and heat compression in an embodiment of a method for producing a fiber-reinforced molded body of the third aspect.
- (17) FIG. 17 is a cross-sectional view showing lamination and heat compression in an embodiment of the method for producing a fiber-reinforced molded body of the third aspect.
- (18) FIG. 18 is a graph showing results of viscosity measurement of mixed resins used in Examples 1C and 6C to 10C, and Comparative Examples 1C and 2C according to the third aspect.
- (19) FIG. 19 is a cross-sectional view of a fiber-reinforced sandwich composite according to an embodiment of the fourth aspect.
- (20) FIG. 20 is a cross-sectional view showing lamination and heat compression in an embodiment of a method for producing a fiber-reinforced sandwich composite of the fourth aspect.
- (21) FIG. 21 is a table showing materials, thicknesses, and the like of core components used in Examples and Comparative Examples according to the fourth aspect.

(22) FIG. 22 is a table showing melting start temperatures, curing reaction start temperatures, and the like of thermosetting resins used in Examples and Comparative Examples according to the fourth aspect.

(23) FIG. 23 is a table showing configurations, physical property values, and the like of Examples and Comparative Examples according to the fourth aspect.

(24) FIG. 24 is a graph showing results of viscosity measurement of thermosetting resins used in Examples 1D, 6D, and 7D and Comparative Examples 2D and 3D according to the fourth aspect.

(25) FIG. 25 is a cross-sectional view of a fiber-reinforced molded body according to a first embodiment of the fifth aspect.

(26) FIG. 26 is a cross-sectional view showing lamination and heat compression in an embodiment of a method for producing the fiber-reinforced molded body in FIG. 25 of the fifth aspect.

(27) FIG. 27 is a cross-sectional view showing lamination and heat compression in another embodiment of the method for producing the fiber-reinforced molded body in FIG. 25 of the fifth aspect.

(28) FIG. 28 is a cross-sectional view of a fiber-reinforced molded body according to a second embodiment of the fifth aspect.

(29) FIG. 29 is a cross-sectional view showing lamination and heat compression in an embodiment of a method for producing the fiber-reinforced molded body in FIG. 28.

(30) FIG. 30 is a cross-sectional view of a fiber-reinforced molded body according to a third embodiment of the fifth aspect.

(31) FIG. 31 is a cross-sectional view showing lamination and heat compression in an embodiment of a method for producing the fiber-reinforced molded body in FIG. 30.

(32) FIG. 32 is a cross-sectional view of a fiber-reinforced molded body according to a fourth embodiment of the fifth aspect.

(33) FIG. 33 is a cross-sectional view showing lamination and heat compression in an embodiment of a method for producing the fiber-reinforced molded body in FIG. 32 of the fifth aspect.

(34) FIG. 34 is a cross-sectional view showing lamination and heat compression in another embodiment of the method for producing the fiber-reinforced molded body in FIG. 32 of the fifth aspect.

(35) FIG. 35 is a table showing types, melting start temperatures, curing reaction start temperatures, and the like of powdery thermosetting resins used in Examples and Comparative Examples according to the fifth aspect.

(36) FIG. 36 is a table, split across multiple sheets labeled as views FIGS. 36A-36C because it is too large to fit on a single sheet within the margins, showing configurations, physical property values, and the like of Examples and Comparative Examples according to the fifth aspect.

(37) FIG. 37 is a graph showing results of viscosity measurement of thermosetting resins used in Examples 1E, 6E to 15E, and 20E to 26E and Comparative Examples 1E to 4E according to the fifth aspect.

DESCRIPTION OF EMBODIMENTS

(38) Hereinafter, the present disclosure will be described in detail. In the present specification, the description using “to” for the numerical range includes the lower limit and the upper limit unless otherwise specified. For example, the expression “10 to 20” includes both the lower limit value “10” and the upper limit value “20”. That is, the expression “10 to 20” has the same meaning as “10 or more and 20 or less”.

First Aspect

(39) An embodiment of the first aspect will be described. In a fiber-reinforced resin molded body **10** of an embodiment shown in FIG. 1, a plurality of laminated fiber substrates **11** are heat-compressed together with a thermosetting resin **15**, and shaped and integrated by curing of the thermosetting resin **15**.

(40) The fiber substrate **11** may be a single layer or a plurality of layers, and the number of layers is

determined according to, for example, the use of the fiber-reinforced resin molded body **10**. In the illustrated form, the fiber substrate **11** is composed of four layers. Examples of the fiber substrate **11** include woven fabrics and nonwoven fabrics made of glass fibers, aramid fibers, basalt fibers, and carbon fibers, but are not particularly limited thereto. The fiber substrate **11** is preferably a carbon fiber woven fabric because it is excellent in lightweight and high rigidity. The carbon fiber woven fabric is preferably a woven fabric in which fibers are not unidirectional. Preferred examples thereof include plain weave, twill weave, satin weave, all of which are constituted of warp and weft, and triaxial weave which is constituted of threads in three directions. The carbon fiber woven fabric preferably has a fiber weight of 50 to 600 g/m² from the viewpoint of impregnation of the thermosetting resin **15** and rigidity of the fiber-reinforced resin molded body **10**.

(41) As the thermosetting resin **15**, a solid powder is used in the production of the fiber-reinforced resin molded body **10**. The shape of the powder is not particularly limited, and examples thereof include a spherical shape, a needle shape, and a flake shape. The thermosetting resin powder is disposed in contact with the fiber substrate **11**, and when the fiber substrate **11** is heat-compressed together with the thermosetting resin powder, the thermosetting resin powder is melted, impregnated into the fiber substrate **11**, and cured. As a form of disposing the thermosetting resin powder in contact with the fiber substrate **11**, the following forms are exemplified. When the fiber substrate **11** is a single layer, the thermosetting resin powder is disposed on at least one of the upper surface and the lower surface of the single layer of the fiber substrate **11**, and when the fiber substrate **11** is a plurality of layers, the thermosetting resin powder is disposed on at least one surface, that is, at least one surface among the uppermost surface, the lowermost surface, and the lamination surface (between fiber substrates) of the plurality of layers.

(42) The thermosetting resin **15** preferably satisfies $30 \leq (T_b - T_a) \leq 100$, and more preferably $40 \leq (T_b - T_a) \leq 70$, when the melting start temperature is $T_a^\circ \text{C.}$ and the curing reaction start temperature is $T_b^\circ \text{C.}$ When the value of $(T_b - T_a)$ is in this range, the melted thermosetting resin **15** can be sufficiently impregnated into the fiber substrate **11**, so that the fiber-reinforced resin molded body **10** having uniform physical properties can be obtained.

(43) The thermosetting resin **15** has a minimum viscosity of preferably 2,000 Pa·s or less, more preferably 1,500 Pa·s or less at a melting start temperature of $T_a^\circ \text{C.}$ or higher. When the minimum viscosity is in this range, the melted thermosetting resin **15** can be sufficiently impregnated into the fiber substrate **11**, so that the fiber-reinforced resin molded body **10** having uniform physical properties can be obtained.

(44) The thermosetting resin **15** preferably has a maximum viscosity of 1,000 Pa·s or more in a temperature range of the curing reaction start temperature $T_b^\circ \text{C.}$ to 190°C. When the maximum viscosity is in this range, the melted thermosetting resin **15** can be impregnated into the fiber substrate **11** and retained therein. As a result, the shapability of the fiber-reinforced resin molded body **10** is good, and sufficient strength can be obtained in a short time.

(45) The thermosetting resin **15** preferably has a melting start temperature $T_a^\circ \text{C.}$ of 60 to 100°C. When the melting start temperature $T_a^\circ \text{C.}$ of the thermosetting resin **15** is in this range, temperature control can be easily performed when the laminate in which a powder of the thermosetting resin **15** is disposed between at least any two adjacent fiber substrates **11** is heat-compressed to melt and cure the thermosetting resin **15**.

(46) The thermosetting resin that can satisfy the melting start temperature $T_a^\circ \text{C.}$, the curing reaction start temperature $T_b^\circ \text{C.}$, the range of the value of $(T_b - T_a)$, the minimum viscosity, and the maximum viscosity as described above is preferably selected from the group consisting of a phenol resin, a mixed resin of a phenol resin and an epoxy resin, a mixed resin of a phenol resin and a cyanate resin, and a mixed resin of a phenol resin, a cyanate resin, and an epoxy resin. The phenol resin is excellent in flame retardancy, and thus can impart excellent strength and flame retardancy to the fiber-reinforced resin molded body **10**.

(47) Various powder additives such as a pigment, an antibacterial agent, and an ultraviolet absorber may be added to the thermosetting resin **15** as long as the viscosity and reactivity of the thermosetting resin are not affected.

(48) The method for producing a fiber-reinforced resin molded body according to the present aspect is performed by disposing a thermosetting resin powder in contact with a fiber substrate; and heat-compressing the fiber substrate together with the thermosetting resin powder by a mold so that the thermosetting resin powder is melted, impregnated into the fiber substrate, and cured. The fiber substrate and the thermosetting resin are as described above for the fiber-reinforced resin molded body **10**.

(49) The form of disposing the thermosetting resin powder is as described above, i.e., when the fiber substrate is a single layer, the thermosetting resin powder is disposed on at least one of the upper surface and the lower surface of the fiber substrate, and when the fiber substrate is a plurality of layers, the thermosetting resin powder is disposed on at least one of the uppermost surface, the lowermost surface, and the lamination surface (between fiber substrates) of a plurality of layers.

(50) When the thermosetting resin powder is disposed on the lamination surface (between fiber substrates) of a plurality of layers of the fiber substrate, the thermosetting resin powder may be disposed on not only one lamination surface (between two adjacent fiber substrates), but also on all lamination surfaces (between all fiber substrates) or on every predetermined number of lamination surfaces (between every predetermined number of fiber substrates). The position of the surface where the powder is to be disposed and the number of surfaces where the powder is to be disposed are appropriately determined according to, for example, the number of laminated fiber substrates.

(51) When the thermosetting resin powder is disposed in contact with the upper surface or the lower surface of the single layer of the fiber substrate or the uppermost surface or the lowermost surface of the plurality of layers of the fiber substrate, a mold release paper may be disposed between the thermosetting resin powder and the mold surface of the mold for convenience of operation.

(52) An embodiment of a method for producing the fiber-reinforced resin molded body **10** including the fiber substrate **11** composed of four layers as shown in FIG. **1**, will be described with reference to FIG. **2**. In the following description of the production method, the plurality of fiber substrates **11** are denoted by a combined reference numeral of “**11**” and “alphabet”, such as “**11A**”, in order to easily grasp the vertical positional relationship in the plurality of fiber substrates **11**.

(53) In the embodiment shown in FIG. **2**, when four fiber substrates **11A** to **11D** are laminated, a thermosetting resin powder **15A** is disposed between lower two fiber substrates **11A** and **11B** and upper two fiber substrates **11C** and **11D** (between the fiber substrate **11B** and the fiber substrate **11C**). The particle size of the thermosetting resin powder **15A** is preferably 10 to 500 μm from the viewpoint of ease of melting. The amount of the thermosetting resin powder **15A** is preferably adjusted so that the VF value (%) of the molded body is 40 to 70%. The VF value (%) is a value calculated by $(\text{total weight of fiber substrate} / \text{density of fiber}) / (\text{volume of molded body}) \times 100$.

(54) A laminate of the fiber substrates **11A** to **11D**, the laminate being prepared by disposing the thermosetting resin powder **15A** between the fiber substrates **11B** and **11C** and laminating the fiber substrates, is sandwiched between a lower mold **31** and an upper mold **32** of a mold **30** that has been heated, and then the laminate is heat-compressed. The mold **30** is heated by a heating unit such as an electric heater to a temperature $T_c^\circ \text{C.}$ at which the thermosetting resin powder **15A** can be melted and cured.

(55) The temperature $T_c^\circ \text{C.}$ of the mold **30** is set to a temperature satisfying preferably $[T_b + (T_b - T_a)/3] - 15 \leq T_c \leq [T_b + (T_b - T_a)/3] + 20$, more preferably $[T_b + (T_b - T_a)/3] - 10 \leq T_c \leq [T_b + (T_b - T_a)/3] + 20$ with respect to the melting start temperature $T_a^\circ \text{C.}$ and the curing reaction start temperature $T_b^\circ \text{C.}$ of the thermosetting resin. For example, when $T_a^\circ \text{C.}$ is 70°C. and $T_b^\circ \text{C.}$ is 130°C. , $T_c^\circ \text{C.}$ is 140°C. to 170°C.

(56) Pressurization (compression) of the fiber substrates **11A** to **11D** during heat compression by

the mold **30** is preferably 2 to 20 MPa so that the melted thermosetting resin of the thermosetting resin powder **15A** between the fiber substrates can be satisfactorily impregnated into the fiber substrates **11A** to **11D**.

(57) The compression percentage (%) of the fiber substrates **11A** to **11D** is a value calculated as (distance between mold surface of lower mold **31** and mold surface of upper mold **32**)/(total thickness of all layers of fiber substrate)×100, and is preferably 60 to 100%.

(58) The thermosetting resin powder **15A** between the fiber substrates (between the fiber substrate **11B** and the fiber substrate **11C**) is melted through heating of the laminate by the mold **30**, and the melted thermosetting resin is impregnated into the fiber substrates **11B** and **11A** on the lower side and the fiber substrates **11C** and **11D** on the upper side through compression of the laminate. Then, the thermosetting resin that has been impregnated into the fiber substrates **11A** to **11D** is cured, whereby the fiber substrates **11A** to **11D** are integrated in a compressed state. Thus, the fiber-reinforced resin molded body **10** of FIG. **1**, which is shaped into the shape of the mold surface of the lower mold **31** and the upper mold **32**, is obtained.

(59) FIG. **3** shows an embodiment in which six fiber substrates **11A** to **11F** are laminated, thermosetting resin powders **15A** to **15E** are disposed between all the fiber substrates, and the laminate is heat-compressed by the mold **30**.

(60) The amount (total amount) of the thermosetting resin powder **15A**, the heating temperature of the mold **30**, the pressurization of the laminate, and the like are as described in the embodiment of FIG. **2**.

Examples

(61) Fiber-reinforced resin molded bodies of Examples 1A to 7A and Comparative Examples 1A and 2A were prepared as follows using the thermosetting resins shown in FIG. **4**. The viscosity of the thermosetting resin was measured under the following conditions using a rheometer Rheosol-G3000, manufactured by UBM. 1) An amount of 0.4 g of a sample is formed into pellets (diameter ϕ : 18 mm, thickness: about 0.4 mm), and the formed pellets are sandwiched between parallel plates having a diameter ϕ of 18 mm. 2) The dynamic viscosity was measured at 2° C. intervals over a temperature range of 40° C. to 200° C. at a temperature rising rate of 5° C./min, a frequency of 1 Hz, and a rotation angle (strain) of 0.1 deg under constant temperature rising.

Preparation of Example 1A

(62) As the fiber substrate, four sheets of carbon fiber woven fabric (manufactured by Teijin Limited, product name: W-3101, weight per unit area: 200 g/m², thickness: 0.22 mm) each cut into 250×200 mm were prepared. The weight of the cut fiber substrate was 10 g per sheet. Two fiber substrates were laminated, 25 g of a resin A1 as the thermosetting resin powder was disposed thereon substantially uniformly, and then the remaining two fiber substrates were laminated thereon to prepare a pre-molding laminate.

(63) The resin A1 is a phenol resin (manufactured by Sumitomo Bakelite Co., Ltd., product name: PR-50252) having an average particle size of 30 μ m, a melting start temperature Ta of 69° C., a reaction start temperature Tb of 135° C., a value of (Tb-Ta) of 66° C., a minimum viscosity (minimum viscosity at a temperature of the melting start temperature Ta° C. or higher) of 59 Pa·Math.s, a maximum viscosity (maximum viscosity in a temperature range of the curing reaction start temperature Tb° C. to 190° C.) of 8,768 Pa·Math.s, a value of (Tb-Ta)/3 of 22° C., and a value of Tb+(Tb-Ta)/3 of 157° C.

(64) The results of viscosity measurement of the resin A1 (Examples 1A, 2A, 3A) are shown in the graph of FIG. **6**.

(65) The pre-molding laminate was disposed on a molding surface (mold surface) of a lower mold of a mold heated to 150° C., and an upper mold of the mold was placed on the pre-molding laminate to close the mold. The laminate was heat-compressed at a pressure of 5 MPa and 150° C. for 10 minutes. The thermosetting resin powder disposed between the fiber substrates at the center was melted by heating, and impregnated into the respective layers of the fiber substrates by

compression of the pre-molding laminate, and in this state, curing of the thermosetting resin was completed. Thus, a fiber-reinforced resin molded body of Example 1A composed of an integrated laminate product in which the thermosetting resin was impregnated into four layers of fiber substrates and cured, was prepared.

Preparation of Example 2A

(66) A fiber-reinforced resin molded body of Example 2A was prepared by heat-compressing under the same conditions as in Example 1A except that the same four fiber substrates and resin A1 as in Example 1A were used, and 8.3 g of the resin A1 (the total amount of the resin A1 between all the substrates: 24.9 g) was disposed between the respective fiber substrates (between all the fiber substrates) substantially uniformly to prepare a pre-molding laminate.

Preparation of Example 3A

(67) A fiber-reinforced resin molded body of Example 3A was prepared by heat-compressing under the same conditions as in Example 1A except that the same ten fiber substrates as in Example 1A were prepared, five fiber substrates were laminated, 60 g of the resin A was disposed thereon substantially uniformly, and then the remaining five fiber substrates were laminated thereon to prepare a pre-molding laminate.

Preparation of Example 4A

(68) A fiber-reinforced resin molded body of Example 4A was prepared in the same manner as in Example 1A except that the resin B1 was used as the thermosetting resin.

(69) The resin B1 is a phenol resin (manufactured by Sumitomo Bakelite Co., Ltd., product name: PR-310) having an average particle size of 30 μm , a melting start temperature T_a of 95° C., a reaction start temperature T_b of 135° C., a value of $(T_b - T_a)$ of 40° C., a minimum viscosity (minimum viscosity at a temperature of the melting start temperature T_a ° C. or higher) of 1,500 Pa.Math.s, a maximum viscosity (maximum viscosity in a temperature range of the curing reaction start temperature T_b ° C. to 190° C.) of 209,004 Pa.Math.s, a value of $(T_b - T_a)/3$ of 13° C., and a value of $T_b + (T_b - T_a)/3$ of 148° C.

Preparation of Example 5A

(70) A fiber-reinforced resin molded body of Example 5A was prepared in the same manner as in Example 1A except that a resin (25 g) prepared by uniformly mixing 12.5 g of the resin A1 and 12.5 g of a resin C1 was used as the thermosetting resin, and the mold temperature was 170° C.

(71) As the resin C1, an epoxy resin (manufactured by Mitsubishi Chemical Corporation, product name: jER-1001) was pulverized in a mortar and used. The average particle size was 100 μm .

(72) The resin prepared by uniformly mixing 12.5 g of the resin A1 and 12.5 g of the resin C1 has a melting start temperature T_a of 73° C., a reaction start temperature T_b of 140° C., a value of $(T_b - T_a)$ of 67° C., a minimum viscosity (minimum viscosity at a temperature of the melting start temperature T_a ° C. or higher) of 22 Pa.Math.s, a maximum viscosity (maximum viscosity in a temperature range of the curing reaction start temperature T_b ° C. to 190° C.) of 5,180 Pa.Math.s, a value of $(T_b - T_a)/3$ of 22° C., and a value of $T_b + (T_b - T_a)/3$ of 162° C.

Preparation of Example 6A

(73) A fiber-reinforced resin molded body of Example 6A was prepared in the same manner as in Example 1A except that a resin (25 g) prepared by uniformly mixing 12.5 g of a resin D1 and 12.5 g of a resin E1 was used as the thermosetting resin, and the mold temperature was 160° C.

(74) As the resin D1, a phenol resin (manufactured by Sumitomo Bakelite Co., Ltd., product name: PR-50235D) was pulverized in a mortar and used. The average particle size was 90 μm .

(75) As the resin E1, a cyanate resin (manufactured by Mitsubishi Gas Chemical Company, Inc., product name: CYTESTER TA) was pulverized in a mortar and used. The average particle size was 100 μm .

(76) The resin prepared by uniformly mixing 12.5 g of the resin D1 and 12.5 g of the resin E1 has a melting start temperature T_a of 76° C., a reaction start temperature T_b of 138° C., a value of $(T_b - T_a)$ of 62° C., a minimum viscosity (minimum viscosity at a temperature of the melting start

temperature T_a ° C. or higher) of 475 Pa.Math.s, a maximum viscosity (maximum viscosity in a temperature range of the curing reaction start temperature T_b ° C. to 190° C.) of 51,895 Pa.Math.s, a value of $(T_b - T_a)/3$ of 21° C., and a value of $T_b + (T_b - T_a)/3$ of 159° C.

Preparation of Example 7A

(77) A fiber-reinforced resin molded body of Example 7A was prepared in the same manner as in Example 1A except that a resin (24.9 g) prepared by uniformly mixing 8.3 g of the resin D1, 8.3 g of the resin E1, and 8.3 g of the resin C1 was used as the thermosetting resin, and the mold temperature was 170° C.

(78) The resin prepared by uniformly mixing 8.3 g of the resin D1, 8.3 g of the resin E1, and 8.3 g of the resin C1 has a melting start temperature T_a of 75° C., a reaction start temperature T_b of 139° C., a value of $(T_b - T_a)$ of 64° C., a minimum viscosity (minimum viscosity at a temperature of the melting start temperature T_a ° C. or higher) of 575 Pa.Math.s, a maximum viscosity (maximum viscosity in a temperature range of the curing reaction start temperature T_b ° C. to 190° C.) of 19,025 Pa.Math.s, a value of $(T_b - T_a)/3$ of 21° C., and a value of $T_b + (T_b - T_a)/3$ of 160° C.

Preparation of Comparative Example 1A

(79) A fiber-reinforced resin molded body of Comparative Example 1A was prepared in the same manner as in Example 1A except that a resin F1 was used as the thermosetting resin and the mold temperature was 100° C.

(80) The resin F1 is a phenol resin (manufactured by Sumitomo Bakelite Co., Ltd., product name: PR-50699) having an average particle size of 30 μm , a melting start temperature T_a of 72° C., a reaction start temperature T_b of 91° C., a value of $(T_b - T_a)$ of 19° C., a minimum viscosity (minimum viscosity at a temperature of the melting start temperature T_a ° C. or higher) of 118,908 Pa.Math.s, a maximum viscosity (maximum viscosity in a temperature range of the curing reaction start temperature T_b ° C. to 190° C.) of 164,468 Pa.Math.s, a value of $(T_b - T_a)/3$ of 6° C., and a value of $T_b + (T_b - T_a)/3$ of 97° C.

(81) The results of viscosity measurement of the resin F1 (Comparative Example 1A) are shown in the graph of FIG. 6.

(82) In Comparative Example 1A, the minimum viscosity and the maximum viscosity of the resin F1 used were high, and the impregnation property of the resin F1 into the fiber substrate (carbon fiber woven fabric) was poor. As a result, a molded body uniformly impregnated with the thermosetting resin was not obtained.

Preparation of Comparative Example 2A

(83) A fiber-reinforced resin molded body of Comparative Example 2A was prepared in the same manner as in Example 1A except that a resin (24.9 g) prepared by uniformly mixing 8.3 g of the resin A1 and 16.6 g of the resin D1 was used as the thermosetting resin, and the mold temperature was 160° C.

(84) The resin prepared by uniformly mixing 8.3 g of the resin A1 and 16.6 g of the resin D1 has a melting start temperature T_a of 80° C., a reaction start temperature T_b of 140° C., a value of $(T_b - T_a)$ of 60° C., a minimum viscosity (minimum viscosity at a temperature of the melting start temperature T_a ° C. or higher) of 21 Pa.Math.s, a maximum viscosity (maximum viscosity in a temperature range of the curing reaction start temperature T_b ° C. to 190° C.) of 260 Pa.Math.s, a value of $(T_b - T_a)/3$ of 20° C., and a value of $T_b + (T_b - T_a)/3$ of 160° C.

(85) The results of viscosity measurement of the mixed resin of the resin A1 and the resin D1 (weight ratio: 1/2) (Comparative Example 2A) are shown in the graph of FIG. 6.

(86) In Comparative Example 2A, the thermosetting resin was not sufficiently cured, and deformation occurred at the time of demolding the molded body. As a result, good molded body was not obtained.

(87) For the obtained fiber-reinforced resin molded bodies of Examples 1A to 7A and Comparative Examples 1A and 2A, the thickness (mm), the VF value (%), the flexural strength (MPa), and the flexural modulus (GPa) were measured and the appearance was evaluated. The results are as shown

in FIG. 5 and will be described below.

(88) The VF value (%) was calculated as (total weight of fiber substrate/density of fiber)/(volume of molded body)×100.

(89) The flexural strength (MPa) and the flexural modulus (GPa) were measured based on the method according to JIS K7074 A.

(90) In the evaluation of the appearance, presence of a defect such as deformation or non-uniformity of impregnation of resin on the surface of the molded body was visually confirmed, and the case where there was no defect was evaluated as “○”, and the case where there was a defect was evaluated as “×”.

Physical Properties and the Like of Example 1A

(91) The fiber-reinforced resin molded body of Example 1A was prepared by disposing a thermosetting resin powder between fiber substrates at the center, with the number of laminated fiber substrates being four and the thermosetting resin being the resin A1. The molded body had a thickness of 0.8 mm, a VF value of 58%, a flexural strength of 550 MPa, a flexural modulus of 54 GPa, and an appearance of “○”, and thus had high strength and rigidity, and a good appearance.

Physical Properties and the Like of Example 2A

(92) The fiber-reinforced resin molded body of Example 2A was prepared by disposing a thermosetting resin powder between respective fiber substrates, with the number of laminated fiber substrates being four and the thermosetting resin being the resin A1. The molded body had a thickness of 0.8 mm, a VF value of 58%, a flexural strength of 600 MPa, a flexural modulus of 55 GPa, and an appearance of “○”. Example 2A was prepared by disposing the thermosetting resin powder between the respective fiber substrates, and accordingly, Example 2A had a higher strength and rigidity than those of Example 1A.

Physical Properties and the Like of Example 3A

(93) The fiber-reinforced resin molded body of Example 3A was prepared by disposing a thermosetting resin powder between fiber substrates at the center, with the number of laminated fiber substrates being ten and the thermosetting resin being the resin A1. The molded body had a thickness of 2.0 mm, a VF value of 59%, a flexural strength of 620 MPa, and a flexural modulus of 53 GPa, and an appearance of “○”, and thus had high strength and rigidity, and a good appearance.

Physical Properties and the Like of Example 4A

(94) The fiber-reinforced resin molded body of Example 4A was prepared by disposing a thermosetting resin powder between fiber substrates at the center, with the number of laminated fiber substrates being four and the thermosetting resin being the resin B1. The molded body had a thickness of 0.8 mm, a VF value of 55%, a flexural strength of 450 MPa, and a flexural modulus of 45 GPa, and an appearance of “○”, and thus had high strength and rigidity, and a good appearance. Example 4A, which used the resin B1, had lower strength and rigidity than that of Example 1A which used the resin A1.

Physical Properties and the Like of Example 5A

(95) The fiber-reinforced resin molded body of Example 5A was prepared by disposing a thermosetting resin powder between fiber substrates at the center, with the number of laminated fiber substrates being four and the thermosetting resin being a mixed resin of the resin A1 and the resin C1 (weight ratio: 1/1). The molded body had a thickness of 0.8 mm, a VF value of 57%, a flexural strength of 990 MPa, and a flexural modulus of 60 GPa, and an appearance of “○”, and thus had high strength and rigidity, and a good appearance. Example 5A, which used the mixed resin of the resin A1 and the resin C1 (weight ratio: 1/1), had higher strength and rigidity than those of Example 1A which used the resin A1 and Example 4A which used the resin B1.

Physical Properties and the Like of Example 6A

(96) The fiber-reinforced resin molded body of Example 6A was prepared by disposing a thermosetting resin powder between fiber substrates at the center, with the number of laminated fiber substrates being four and the thermosetting resin being a mixed resin of the resin D1 and the

resin E1 (weight ratio: 1/1). The molded body had a thickness of 0.8 mm, a VF value of 59%, a flexural strength of 900 MPa, and a flexural modulus of 61 GPa, and an appearance of “○”, and thus had high strength and rigidity, and a good appearance. Example 6A, which used the mixed resin of the resin D1 and the resin E1 (weight ratio: 1/1), had high strength and rigidity equivalent to those of Example 5A which used the mixed resin of the resin A1 and the resin C1 (weight ratio: 1/1).

Physical Properties and the Like of Example 7A

(97) The fiber-reinforced resin molded body of Example 7A was prepared by disposing a thermosetting resin powder between fiber substrates at the center, with the number of laminated fiber substrates being four and the thermosetting resin being a mixed resin of the resin D1, the resin E1, and the resin C1 (weight ratio: 1/1/1). The molded body had a thickness of 0.8 mm, a VF value of 58%, a flexural strength of 930 MPa, and a flexural modulus of 58 GPa, and an appearance of “○”, and thus had high strength and rigidity, and a good appearance. Example 7A, which used the mixed resin of the resin D1, the resin E1, the resin C1 (weight ratio: 1/1/1), had high strength and rigidity equivalent to those of Example 5A which used the mixed resin of the resin A1 and the resin C1 (weight ratio: 1/1) and Example 6A which used the mixed resin of the resin D1 and the resin E1 (weight ratio: 1/1).

Physical Properties and the Like of Comparative Example 1A

(98) The fiber-reinforced resin molded body of Comparative Example 1A was prepared by disposing a thermosetting resin powder between fiber substrates at the center, with the number of laminated fiber substrates being four and the thermosetting resin being the resin F1. In this case, the impregnation property of the resin F1 was poor, and a molded body uniformly impregnated with the thermosetting resin was not obtained. Therefore, the thickness, VF value, flexural strength, and flexural modulus could not be measured.

Physical Properties and the Like of Comparative Example 2A

(99) The fiber-reinforced resin molded body of Comparative Example 2A was prepared by disposing a thermosetting resin powder between the fiber substrates at the center, with the number of laminated fiber substrates being four and the thermosetting resin being a mixed resin of the resin A1 and the resin D1 (weight ratio: 1/2). In this case, the thermosetting resin was not sufficiently cured, and deformation occurred at the time of demolding the molded body. Therefore, the thickness, VF value, flexural strength, and flexural modulus could not be measured.

(100) In Examples, only the flat plate shape is shown as the shape of the fiber-reinforced resin molded body, but in the present disclosure, the shape of the fiber-reinforced resin molded body (the shape of the mold) is not limited to the flat plate shape, and may be any shape such as a curved shape or an irregularity shape.

(101) As described above, according to the present aspect, it is possible to obtain a fiber-reinforced resin molded body easily and inexpensively without using a prepreg and safely without a risk of environmental pollution.

Second Aspect

(102) An embodiment of the second aspect will be described.

(103) In a fiber-reinforced resin molding prepreg **110** of the first embodiment shown in (1-1) of FIG. 7, a fiber substrate **111** is heat-compressed together with a thermosetting resin **115**, and the thermosetting resin **115** is in a state of being adhering to the fiber substrate **111**. The thermosetting resin **115** adhering to the fiber substrate **111** is in a solid state before the start of the curing reaction.

(104) The fiber substrate **111** is composed of one layer in the prepreg **110** of the first embodiment, but may be composed of a plurality of layers. Examples of the fiber substrate **111** include woven fabrics and nonwoven fabrics made of glass fibers, aramid fibers, basalt fibers, and carbon fibers, but are not particularly limited thereto. The fiber substrate **111** is preferably a carbon fiber woven fabric because it is excellent in lightweight and high rigidity. The carbon fiber woven fabric is preferably a woven fabric in which fibers are not unidirectional. Preferred examples thereof include

plain weave, twill weave, satin weave, all of which are constituted of warp and weft, and triaxial weave which is constituted of threads in three directions. The carbon fiber woven fabric preferably has a fiber weight of 50 to 600 g/m.² from the viewpoint of impregnation of the thermosetting resin **115** and rigidity of the fiber-reinforced resin molded body.

(105) As the thermosetting resin **115**, one that is in a solid powder form before heat compression is used in the preparation of the prepreg **110**. The shape of the powder is not particularly limited, and examples thereof include a spherical shape, a needle shape, and a flake shape.

(106) In the preparation of the prepreg **110**, the powder of the thermosetting resin **115** is disposed in contact with the fiber substrate **111**, and when the fiber substrate **111** is heat-compressed together with the powder of the thermosetting resin **115**, the powder of the thermosetting resin is melted and impregnated into the fiber substrate **111**, and then cooled and solidified in a state before the start of the curing reaction.

(107) The thermosetting resin **115** has a viscosity of 2,000 Pa·s or less, preferably 1,500 Pa·s or less at the curing reaction start temperature T_b ° C.

(108) With the viscosity of 2,000 Pa·s or less at the curing reaction start temperature T_b ° C., the thermosetting resin **115** can be uniformly impregnated into the fiber substrate **111** when a fiber-reinforced resin molded body is produced using the prepreg **110**. As a result, a fiber-reinforced resin molded body with good quality is obtained.

(109) The thermosetting resin **115** has a maximum viscosity of 1,000 Pa·s or more in a temperature range of the curing reaction start temperature T_b ° C. to 190° C.

(110) With the maximum viscosity of 1,000 Pa·s or more in a temperature range of the curing reaction start temperature T_b ° C. to 190° C., the melted thermosetting resin **115** can be impregnated into the fiber substrate **111** and sufficiently cured when a fiber-reinforced resin molded body is produced using the prepreg **110**. As a result, the shapability of the fiber-reinforced resin molded body is good, and sufficient strength can be obtained in a short time (about 10 minutes).

(111) The thermosetting resin **115** preferably has a melting start temperature T_a ° C. of 60 to 100° C. When the melting start temperature T_a ° C. is in the above range, the heating temperature in the preparation of the prepreg **110** does not need to be so high, which facilitates the preparation of the prepreg **110**.

(112) The melting start temperature T_a ° C. and the curing reaction start temperature T_b ° C. of the thermosetting resin **115** preferably satisfy $30 \leq (T_b - T_a) \leq 100$, and more preferably satisfy $40 \leq (T_b - T_a) \leq 70$. When the value of $(T_b - T_a)$ is in this range, the thermosetting resin **115** is sufficiently impregnated into the fiber substrate **111** when a fiber-reinforced resin molded body is produced using the prepreg **110**. As a result, a fiber-reinforced resin molded body having uniform physical properties can be obtained.

(113) In the thermosetting resin **115**, it is preferable that, with respect to T_c ° C. which is a temperature during heat compression in the preparation of the prepreg **110**, the melting start temperature T_a ° C. be equal to or lower than T_c ° C., and the curing reaction start temperature T_b ° C. be equal to or higher than T_c ° C. More preferably, T_a ° C. is T_c ° C. minus 5° C. or lower, and T_b ° C. is T_c ° C. plus 5° C. or higher. When T_a ° C. and T_b ° C. are in these ranges, the thermosetting resin **115** can be melted and impregnated into the fiber substrate **111** in the preparation of the prepreg **110**, and the curing reaction of the thermosetting resin **115** does not start in the preparation of the prepreg **110**. Therefore, the quality of the prepreg **110** is improved and the storage stability of the prepreg **110** is improved.

(114) The particle size of the powder of the thermosetting resin **115** used in the preparation of the prepreg **110** is preferably 10 to 500 μm from the viewpoint of ease of melting. The amount of the thermosetting resin **115** used in the preparation of the prepreg is preferably adjusted so that the VF value (%) of the fiber-reinforced resin molded body is 40 to 70%. The VF value (%) is a value calculated by $(\text{total weight of fiber substrate} / \text{density of fiber}) / (\text{volume of fiber-reinforced resin molded body}) \times 100$.

(115) The thermosetting resin that can satisfy the melting start temperature $T_a^{\circ}\text{C.}$, the curing reaction start temperature $T_b^{\circ}\text{C.}$, the range of the value of $(T_b - T_a)$, the minimum viscosity, the maximum viscosity, and the like as described above is preferably selected from the group consisting of a phenol resin, a mixed resin of a phenol resin and an epoxy resin, a mixed resin of a phenol resin and a cyanate resin, and a mixed resin of a phenol resin, a cyanate resin, and an epoxy resin. The phenol resin is excellent in flame retardancy, and thus can impart excellent strength and flame retardancy to the fiber-reinforced resin molded body.

(116) Various powder additives such as a pigment, an antibacterial agent, and an ultraviolet absorber may be added to the thermosetting resin **115** as long as the viscosity and reactivity of the thermosetting resin are not affected.

(117) One mode of preparation of the fiber-reinforced resin molding prepreg **110** of the first embodiment will be described below. In the following description, the “thermosetting resin powder **115**” may be denoted by a combined reference numeral of number “**115**” and “alphabet”, such as “thermosetting resin powder **115A**”.

(118) As shown in (1-2) of FIG. 7, the fiber substrate **111** and the thermosetting resin powder **115A** disposed in contact with the fiber substrate **111** are sandwiched between plastic mold release films **141A** and **141B**, and these are sandwiched between a lower mold **151** and an upper mold **152** of a heated mold **150**, and heat-compressed to prepare the prepreg **110** of the first embodiment. At this time, the thermosetting resin powder **115A** is disposed on one or both of the upper and lower surfaces of the fiber substrate **111**.

(119) The mold **150** is heated by a heating unit such as an electric heater. The temperature $T_c^{\circ}\text{C.}$ during heat compression (temperature of the mold) in the preparation of the prepreg **110** is equal to or higher than the melting start temperature $T_a^{\circ}\text{C.}$ and equal to lower than the curing reaction start temperature $T_b^{\circ}\text{C.}$ of the thermosetting resin powder **115A** (thermosetting resin **115**). More preferably, the temperature $T_c^{\circ}\text{C.}$ during heat compression (temperature of the mold) is the melting start temperature $T_a^{\circ}\text{C.}$ of the thermosetting resin **115** plus 5°C. or higher, and the curing reaction start temperature $T_b^{\circ}\text{C.}$ minus 5°C. or lower.

(120) Pressurization (compression) by the mold **150** in the preparation of the prepreg is preferably 0.1 to 10 MPa so that the melted thermosetting resin of the thermosetting resin powder **115A** can be satisfactorily impregnated into the fiber substrate **111**.

(121) The compression percentage (%) of the fiber substrate **111** is a value calculated as (distance between mold surface of lower mold **151** and mold surface of upper mold **152** – total thickness of mold release film)/(total thickness of all layers of fiber substrate) $\times 100$. The compression percentage is preferably 60 to 100%.

(122) The thermosetting resin powder **115A** is melted by heat compression and impregnated into the fiber substrate **111**, and then cooled and solidified in a state before the start of the curing reaction.

(123) The method for preparing the prepreg is not limited to the above-described method using the mold, but may be other methods such as a heat roller method in which heat compression is performed by a heat roller, and a double belt method in which heat compression is performed between upper and lower belts.

(124) The second embodiment will be described. In the following description, the fiber substrates may be denoted by a combined reference numeral of number “**111**” and “alphabet”, such as “**111A**”, in order to easily grasp the vertical positional relationship in the plurality of fiber substrates **111**.

(125) A fiber-reinforced resin molding prepreg **120** of the second embodiment shown in (2-1) of FIG. 8 includes four layers of fiber substrates **111A**, **111B**, **111C**, and **111D** and the thermosetting resin **115**, and the thermosetting resin **115** is in a state of being adhering to the fiber substrates **111A** to **111D**. The fiber substrates **111A** to **111D** and the thermosetting resin **115** are as described in the prepreg **110** of the first embodiment.

(126) One mode of preparation of the fiber-reinforced resin molding prepreg **120** of the second embodiment will be described below.

(127) As shown in (2-2) of FIG. **8**, a laminate, in which the thermosetting resin powder **115A** is sandwiched between two fiber substrates **111A** and **111B** on the lower side and two fiber substrates **111C** and **111D** on the upper side, is sandwiched between the plastic mold release films **141A** and **141B**, and these are sandwiched between the lower mold **151** and the upper mold **152** of the heated mold **150**, and heat-compressed to prepare the fiber-reinforced resin molding prepreg **120** of the second embodiment. Heat-compression conditions are as described in the first embodiment. The thermosetting resin powder **115A** is melted by heat compression and impregnated into the fiber substrates **111A** to **111D**, and then cooled and solidified in a state before the start of the curing reaction.

(128) The third embodiment will be described. A fiber-reinforced resin molding prepreg **130** of the third embodiment shown in (3-1) of FIG. **9** includes four layers of the fiber substrates **111A**, **111B**, **111C**, and **111D** and the thermosetting resin **115** as in the second embodiment, and the thermosetting resin **115** is in a state of being adhering to the fiber substrates **111A** to **111D**. In the third embodiment, the disposition of the thermosetting resin powder **115** for preparing the prepreg **130** is different from that of the second embodiment as described below.

(129) One mode of preparation of the fiber-reinforced resin molding prepreg **130** of the third embodiment will be described below.

(130) As shown in (3-2) of FIG. **9**, a laminate, in which thermosetting resin powders **115A**, **115B**, and **115C** are sandwiched between respective fiber substrates (between respective layers) of four fiber substrates **111A** to **111D**, is sandwiched between the plastic mold release films **141A** and **141B**, and these are sandwiched between the lower mold **151** and the upper mold **152** of the heated mold **150**, and heat-compressed to prepare the fiber-reinforced resin molding prepreg **130** of the third embodiment. Heat-compression conditions are as described in the first embodiment. The thermosetting resin powders **115A** to **115C** are melted by heat compression and impregnated into the fiber substrates **111A** to **111D**, and then cooled and solidified in a state before the start of the curing reaction.

(131) The method for producing a fiber-reinforced resin molded body using the fiber-reinforced resin molding prepreg of the present disclosure is performed by heat-compressing a fiber-reinforced resin molding prepreg including the fiber substrate **111** composed of a single layer or a plurality of layers, by a shaping mold to react and cure the thermosetting resin **115**. The shape of the mold surface of the shaping mold is not limited to a flat surface, and may be uneven, curved, or the like depending on the product.

(132) The heating temperature (temperature of the shaping mold) T_d ° C. in the production of the fiber-reinforced resin molded body is set to a temperature satisfying preferably $[T_b + (T_b - T_a)/3] - 15 \leq T_d \leq [T_b + (T_b - T_a)/3] + 20$, more preferably $[T_b + (T_b - T_a)/3] - 10 \leq T_d \leq [T_b + (T_b - T_a)/3] + 20$ with respect to the melting start temperature T_a ° C. and the curing reaction start temperature T_b ° C. of the thermosetting resin **115**. For example, when T_a ° C. is 70° C. and T_b ° C. is 130° C., T_d ° C. is 140° C. to 170° C.

(133) Pressurization (compression) of the fiber-reinforced resin molding prepreg during heat compression by the shaping mold is preferably 2 to 20 MPa so that the melted thermosetting resin **115** can be satisfactorily impregnated into the fiber substrate **111**.

(134) The compression percentage (%) of the fiber-reinforced resin molding prepreg during heat compression by the shaping mold is a value calculated as (distance between mold surface of lower mold **151** and mold surface of upper mold **152**)/(thickness of prepreg)×100, and is preferably 60 to 100%.

(135) The thermosetting resin **115** in the prepreg is melted by heat compression of the fiber-reinforced resin molding prepreg by the shaping mold, and the melted thermosetting resin **115** is reacted and cured, whereby a fiber-reinforced resin molded body shaped into the shape of the mold

surface of the lower mold and the upper mold is obtained.

EXAMPLES

(136) Fiber-reinforced resin molding prepregs of Examples 1B to 7B and Comparative Examples 1B and 2B were prepared by using the thermosetting resin powders shown in FIG. 10, and fiber-reinforced resin molded bodies were prepared using the prepreg immediately after preparation. The viscosity of the thermosetting resin was measured under the following conditions using a rheometer Rheosol-G3000, manufactured by UBM. 1) An amount of 0.4 g of a sample is formed into pellets (diameter ϕ : 18 mm, thickness: about 0.4 mm), and the formed pellets are sandwiched between parallel plates having a diameter ϕ of 18 mm. 2) The dynamic viscosity was measured at 2° C. intervals over a temperature range of 40° C. to 200° C. at a temperature rising rate of 5° C./min, a frequency of 1 Hz, and a rotation angle (strain) of 0.1 deg under constant temperature rising.

Example 1B

(137) Preparation of Prepreg

(138) Example 1B is an example in which the fiber substrate of the prepreg is one sheet (one layer), as shown in (1-1) and (1-2) of FIG. 7. As the fiber substrate, a carbon fiber woven fabric (manufactured by Teijin Limited, product name: W-3101, weight per unit area: 200 g/m², thickness: 0.22 mm) cut into 250×200 mm was prepared. The weight of the cut fiber substrate was 10 g per sheet. The prepared fiber substrate was disposed on a PET film (plastic mold release film) subjected to mold release treatment (applied with a mold release agent), 7 g of the following resin A2 as the thermosetting resin powder was disposed on the fiber substrate substantially uniformly, and a PET film subjected to mold release treatment was placed thereon to prepare a laminate before prepreg molding.

(139) The resin A2 is a phenol resin (manufactured by Sumitomo Bakelite Co., Ltd., product name: PR-50252, average particle size: 30 μ m).

(140) The results of viscosity measurement of the resin A2 (Examples 1B, 2B, 3B) are shown in the graph of FIG. 11.

(141) The laminate before prepreg molding was disposed on a molding surface (mold surface) of a lower mold of a mold heated to 100° C., and an upper mold of the mold was placed on the laminate before prepreg molding to close the mold. The laminate was heat-compressed at a pressure of 0.1 MPa for 5 minutes. The powder of the resin A2 on the fiber substrate was melted and impregnated into the fiber substrate through the heat compression. Then, the mold was opened, the laminate together with the PET films was taken out and naturally cooled to room temperature, and the PET films were removed. Thus, a prepreg of Example 1B was obtained.

(142) Preparation of Fiber-Reinforced Resin Molded Body

(143) A laminate, in which four prepregs of Example 1B were stacked, was disposed on a molding surface (mold surface) of a lower mold of a mold heated to 150° C. in advance, and an upper mold of the mold was placed on the laminate to close the mold. The laminate was heat-compressed at a pressure of 5 MPa for 10 minutes to react and cure the resin A2. Then, the mold was opened and a fiber-reinforced resin molded body of Example 1B was taken out.

Example 2B

(144) Preparation of Prepreg

(145) Example 2B is an example in which the number of fiber substrates of the prepreg is four (four layers), as shown in (2-1) and (2-2) of FIG. 8. A prepreg of Example 2B was prepared in the same manner as in Example 1B except that the same four fiber substrates and resin A2 as in Example 1B were used, and 28 g of the resin A2 was disposed in the middle of the four fiber substrates substantially uniformly.

(146) Preparation of Fiber-Reinforced Molded Body

(147) The prepreg of Example 2B was disposed on a molding surface (mold surface) of a lower mold of a mold heated to 150° C. in advance, and an upper mold of the mold was placed on the laminate to close the mold. The laminate was heat-compressed at a pressure of 5 MPa for 10

minutes to react and cure the resin A2. Then, the mold was opened and a fiber-reinforced resin molded body of Example 2B was taken out.

Example 3B

(148) Preparation of Prepreg

(149) Example 3B is an example in which a prepreg was prepared by disposing the resin A2 between respective layers of four fiber substrates (four layers) of the prepreg, as shown in (3-1) and (3-2) of FIG. 9. A prepreg of Example 3B was prepared in the same manner as in Example 1B except that 9.3 g of the resin A2 was disposed between respective layers of the same four fiber substrates as in Example 1B substantially uniformly.

(150) Preparation of Fiber-Reinforced Molded Body

(151) The prepreg of Example 3B was disposed on a molding surface (mold surface) of a lower mold of a mold heated to 150° C. in advance, and an upper mold of the mold was placed on the laminate to close the mold. The laminate was heat-compressed at a pressure of 5 MPa for 10 minutes to react and cure the resin A2. Then, the mold was opened and a fiber-reinforced resin molded body of Example 3B was taken out.

Example 4B

(152) Preparation of Prepreg

(153) Example 4B is an example in which the fiber substrate of the prepreg is one sheet. A prepreg of Example 4B was prepared in the same manner as in Example 1B except that the following resin B2 was used as the thermosetting resin powder.

(154) The resin B2 is a phenol resin (manufactured by Sumitomo Bakelite Co., Ltd., product name: PR-310, average particle size: 30 μm).

(155) Preparation of Fiber-Reinforced Resin Molded Body

(156) A fiber-reinforced resin molded body of Example 4B was prepared by using four prepregs of Example 4B in the same manner as in Example 1B.

Example 5B

(157) Preparation of Prepreg

(158) Example 5B is an example in which the fiber substrate of the prepreg is one sheet. A prepreg of Example 5B was prepared in the same manner as in Example 1B except that 7 g of a resin prepared by uniformly mixing the resin A2 and the following resin C2 at a weight ratio of 1:1 was used as the thermosetting resin powder.

(159) As the resin C2, an epoxy resin (manufactured by Mitsubishi Chemical Corporation, product name: jER-1001) was pulverized in a mortar and used. The average particle size was 100 μm.

(160) Preparation of Fiber-Reinforced Resin Molded Body

(161) A fiber-reinforced resin molded body of Example 5B was prepared in the same manner as in Example 1B except that four prepregs of Example 5B were used and the temperature of the mold was 170° C.

Example 6B

(162) Preparation of Prepreg

(163) Example 6B is an example in which the fiber substrate of the prepreg is one sheet. A prepreg of Example 6B was prepared in the same manner as in Example 1B except that 7 g of a resin prepared by uniformly mixing the following resin D2 and the following resin E2 at a weight ratio of 1:1 was used as the thermosetting resin powder.

(164) As the resin D2, a phenol resin (manufactured by Sumitomo Bakelite Co., Ltd., product name: PR-50235D) was pulverized in a mortar and used. The average particle size was 90 μm.

(165) As the resin E2, a cyanate resin (manufactured by Mitsubishi Gas Chemical Company, Inc., product name: CYTESTER TA) was pulverized in a mortar and used. The average particle size was 100 μm.

(166) Preparation of Fiber-Reinforced Resin Molded Body

(167) A fiber-reinforced resin molded body of Example 6B was prepared in the same manner as in

Example 1B except that four prepregs of Example 6B were used and the temperature of the mold was 160° C.

Example 7B

(168) Preparation of Prepreg

(169) Example 7B is an example in which the fiber substrate of the prepreg is one sheet. A prepreg of Example 7B was prepared in the same manner as in Example 1B except that 7 g of a resin prepared by uniformly mixing the resin D2, the resin E2, and the resin C2 at a weight ratio of 1:1:1 was used as the thermosetting resin powder.

(170) Preparation of Fiber-Reinforced Resin Molded Body

(171) A fiber-reinforced resin molded body of Example 7B was prepared in the same manner as in Example 1B except that four prepregs of Example 7B were used and the temperature of the mold was 170° C.

Comparative Example 1B

(172) Preparation of Prepreg

(173) A prepreg of Comparative Example 1 was prepared in the same manner as in Example 1B except that 7 g of the following resin F2 was used as the thermosetting resin powder, and the mold temperature was 80° C.

(174) The resin F2 is a phenol resin (manufactured by Sumitomo Bakelite Co., Ltd., product name: PR-50699, average particle size: 30 μm).

(175) The results of viscosity measurement of the resin F2 (Comparative Example 1B) are shown in the graph of FIG. 11.

(176) Preparation of Fiber-Reinforced Resin Molded Body

(177) A fiber-reinforced resin molded body of Comparative Example 1B was prepared in the same manner as in Example 1B except that four prepregs of Comparative Example 1B were used and the temperature of the mold was 100° C.

(178) In Comparative Example 1B, the minimum viscosity and the maximum viscosity of the resin F2 used were high, the impregnation property of the resin F2 into the fiber substrate (carbon fiber woven fabric) was poor. As a result, a fiber-reinforced resin molded body uniformly impregnated with the thermosetting resin was not obtained.

Comparative Example 2B

(179) Preparation of Prepreg

(180) A prepreg of Comparative Example 2B was prepared in the same manner as in Example 1B by using, as the thermosetting resin powder, 7 g of a resin prepared by uniformly mixing the resin A2 and the resin D2 at a weight ratio of 1:2.

(181) The results of viscosity measurement of the resin (Comparative Example 2B) prepared by uniformly mixing the resin A2 and the resin D2 at a weight ratio of 1:2 are shown in the graph of FIG. 11.

(182) Preparation of Fiber-Reinforced Resin Molded Body

(183) A fiber-reinforced resin molded body of Comparative Example 2B was prepared by using four prepregs of Comparative Example 2B in the same manner as in Example 1B.

(184) In Comparative Example 2B, the thermosetting resin used was not sufficiently cured, deformation occurred at the time of demolding the fiber-reinforced resin molded body. As a result, a good molded body was not obtained.

(185) For the fiber-reinforced resin molded bodies of Examples 1B to 7B and Comparative Examples 1B and 2B, the thickness (mm), the VF value (%), the flexural strength (MPa), and the flexural modulus (GPa) were measured and the appearance was evaluated. The results are as shown in FIG. 12 and will be described below.

(186) The flexural strength (MPa) and the flexural modulus (GPa) were measured based on the method according to JIS K7074 A.

(187) In the evaluation of the appearance, presence of a defect such as deformation or non-

uniformity of impregnation of resin on the surface of the molded body was visually confirmed, and the case where there was no defect was evaluated as “○”, and the case where there was a defect was evaluated as “×”.

Physical Properties and the Like of Fiber-Reinforced Resin Molded Body of Example 1B

(188) The fiber-reinforced resin molded body of Example 1B was prepared by laminating four prepregs of Example 1B each prepared by using the resin A2 as the thermosetting resin powder and one fiber substrate. The fiber-reinforced resin molded body of Example 1B had a thickness of 0.8 mm, a VF value of 58%, a flexural strength of 610 MPa, a flexural modulus of 52 GPa, and an appearance of “○”, and thus had high strength and rigidity (flexural modulus) and a good appearance.

Physical Properties and the Like of Fiber-Reinforced Resin Molded Body of Example 2B

(189) The fiber-reinforced resin molded body of Example 2B was prepared from the prepreg of Example 2B prepared by disposing the resin A2 in the middle of four fiber substrates. The fiber-reinforced resin molded body of Example 2B had a thickness of 0.8 mm, a VF value of 58%, a flexural strength of 600 MPa, a flexural modulus of 53 GPa, and an appearance of “○”, and thus had high strength and rigidity, and a good appearance.

Physical Properties and the Like of Fiber-Reinforced Resin Molded Body of Example 3B

(190) The fiber-reinforced resin molded body of Example 3B was prepared from the prepreg of Example 3B prepared by disposing the resin A2 between respective layers of four fiber substrates. The fiber-reinforced resin molded body of Example 3B had a thickness of 0.8 mm, a VF value of 58%, a flexural strength of 620 MPa, a flexural modulus of 53 GPa, and an appearance of “○”, and thus had high strength and rigidity, and a good appearance.

Physical Properties and the Like of Fiber-Reinforced Resin Molded Body of Example 4B

(191) The fiber-reinforced resin molded body of Example 4B was prepared by laminating four prepregs of Example 4B each prepared by using the resin B2 as the thermosetting resin powder and one fiber substrate. The fiber-reinforced resin molded body of Example 4B had a thickness of 0.8 mm, a VF value of 55%, a flexural strength of 460 MPa, a flexural modulus of 46 GPa, and an appearance of “○”, and thus had high strength and rigidity, and a good appearance. Example 4B, which used the resin B2, had lower strength and rigidity than that of Example 1B which used the resin A2.

Physical Properties and the Like of Fiber-Reinforced Resin Molded Body of Example 5B

(192) The fiber-reinforced resin molded body of Example 5B was prepared by laminating four prepregs of Example 5B each prepared by using the mixed resin of the resin A2 and the resin C2 (weight ratio: 1/1) as the thermosetting resin powder, and one fiber substrate. The fiber-reinforced resin molded body of Example 5B had a thickness of 0.8 mm, a VF value of 57%, a flexural strength of 980 MPa, a flexural modulus of 60 GPa, and an appearance of “○”, and thus had high strength and rigidity, and a good appearance. Example 5B, which used the mixed resin of the resin A2 and the resin C2 (weight ratio: 1/1), had higher strength and rigidity than those of Example 1B which used the resin A2 and Example 4B which used the resin B2.

Physical Properties and the Like of Fiber-Reinforced Resin Molded Body of Example 6B

(193) The fiber-reinforced resin molded body of Example 6B was prepared by laminating four prepregs of Example 6B each prepared by using the mixed resin of the resin D2 and the resin E2 (weight ratio: 1/1) as the thermosetting resin powder, and one fiber substrate. The fiber-reinforced resin molded body of Example 6B had a thickness of 0.8 mm, a VF value of 59%, a flexural strength of 910 MPa, a flexural modulus of 61 GPa, and an appearance of “○”, and thus had high strength and rigidity, and a good appearance. Example 6B, which used the mixed resin of the resin D2 and the resin E2 (weight ratio: 1/1), had high strength and rigidity equivalent to those of Example 5B which used the mixed resin of the resin A2 and the resin C2 (weight ratio: 1/1).

Physical Properties and the Like of Fiber-Reinforced Resin Molded Body of Example 7B

(194) The fiber-reinforced resin molded body of Example 7B was prepared by laminating four prepregs of Example 7B each prepared by using the mixed resin of the resin D2, the resin E2, and the resin C2 (weight ratio: 1/1/1) as the thermosetting resin powder and one fiber substrate. The fiber-reinforced resin molded body of Example 7B had a thickness of 0.8 mm, a VF value of 58%, a flexural strength of 930 MPa, a flexural modulus of 59 GPa, and an appearance of “○”, and thus had high strength and rigidity, and a good appearance. Example 7B, which used the mixed resin of the resin D2, the resin E2, and the resin C2 (weight ratio: 1/1/1), had high strength and rigidity equivalent to those of Example 5B which used the mixed resin of the resin A2 and the resin C2 (weight ratio: 1/1) and Example 6B which used the mixed resin of the resin D2 and the resin E2 (weight ratio: 1/1).

Physical Properties and the Like of Fiber-Reinforced Resin Molded Body of Comparative Example 1B

(195) The fiber-reinforced resin molded body of Comparative Example 1B was prepared by laminating four prepregs of Comparative Example 1 each prepared by using the resin F2 as the thermosetting resin powder and one fiber substrate. In this case, the impregnation property of the resin F2 was poor, and a molded body uniformly impregnated with the thermosetting resin was not obtained. Therefore, the thickness, VF value, flexural strength, and flexural modulus could not be measured.

Physical Properties and the Like of Fiber-Reinforced Resin Molded Body of Comparative Example 2B

(196) The fiber-reinforced resin molded body of Comparative Example 2B was prepared by laminating four prepregs of Comparative Example 2B each prepared by using the mixed resin of the resin A2 and the resin D2 (weight ratio: 1/2) as the thermosetting resin powder, and one fiber substrate. In this case, the thermosetting resin was not sufficiently cured and deformation occurred at the time of demolding the molded body. Therefore, the thickness, VF value, flexural strength, and flexural modulus could not be measured.

(197) <Evaluation of Storage Stability of Prepreg>

(198) In order to evaluate the storage stability of the prepregs due to the difference in thermosetting resins, the prepreg of Example 1B including the resin A2, the prepreg of Example 4B including the resin B2, the prepreg of Example 5B including the mixed resin of the resin A2 and the resin C2 (weight ratio: 1/1), the prepreg of Example 6B including the mixed resin of the resin D2 and the resin E2 (weight ratio: 1/1), and the prepreg of Example 7B including the mixed resin of the resin D2, the resin E2, the resin C2 (weight ratio: 1/1/1) were stored in an environment of 23° C. and humidity 50% for 90 days after preparation, and then fiber-reinforced molded bodies were prepared in the same manner as in the corresponding Examples.

(199) For the prepared fiber-reinforced resin molded bodies, evaluation of the appearance of the molded bodies and measurement of the flexural strength and flexural modulus were performed. The results were compared with the results in the case of preparing a fiber-reinforced resin molded body by using a prepreg immediately after the preparation of the prepreg (the results shown in FIG. 12). The results of the storage stability of the prepregs are as shown in FIG. 13. The flexural strength retention in FIG. 13 is a percentage (%) with respect to the value of the flexural strength of a fiber-reinforced resin molded body prepared using a prepreg immediately after preparation (the value of flexural strength shown in FIG. 12). The flexural modulus retention is a percentage (%) with respect to the value of the flexural modulus of a fiber-reinforced resin molded body prepared using a prepreg immediately after preparation (the value of flexural modulus shown in FIG. 12).

(200) The results shown in FIG. 13 for fiber-reinforced resin molded bodies prepared from prepregs after 90 days from the preparation will be described.

Example 1B (Resin A2)

(201) The fiber-reinforced resin molded body of Example 1B, which was prepared 90 days after preparation of the prepreg of Example 1B including the resin A2, had an appearance of “○”, a

flexural strength of 570 MPa, a flexural modulus of 48 GPa, a flexural strength retention of 93%, and a flexural modulus retention of 92%. Thus, the deterioration in physical properties of the fiber-reinforced resin molded body due to storage of the prepreg was small.

Example 4B (Resin B2)

(202) The fiber-reinforced resin molded body of Example 4B, which was prepared 90 days after preparation of the prepreg of Example 4B including the resin B2, had an appearance of “○”, a flexural strength of 420 MPa, a flexural modulus of 42 GPa, a flexural strength retention of 91%, and a flexural modulus retention of 91%. Thus, the deterioration in physical properties of the fiber-reinforced resin molded body due to storage of the prepreg was small.

Example 5B (Resin A2 and Resin C2 at Weight Ratio of 1/1)

(203) The fiber-reinforced resin molded body of Example 5B, which was prepared 90 days after preparation of the prepreg of Example 5B including the mixed resin of the resin A2 and the resin C2 (weight ratio: 1/1), had an appearance of “○”, a flexural strength of 910 MPa, a flexural modulus of 56 GPa, a flexural strength retention of 93%, and a flexural modulus retention of 93%. Thus, the deterioration in physical properties of the fiber-reinforced resin molded body due to storage of the prepreg was small.

Example 6B (Resin D2 and Resin E2 at Weight Ratio of 1/1)

(204) The fiber-reinforced resin molded body of Example 6B, which was prepared 90 days after preparation of the prepreg of Example 6B including the mixed resin of the resin D2 and the resin E2 (weight ratio: 1/1), had an appearance of “○”, a flexural strength of 880 MPa, a flexural modulus of 60 GPa, a flexural strength retention of 97%, and a flexural modulus retention of 98%. Thus, the deterioration in physical properties of the fiber-reinforced resin molded body due to storage of the prepreg was small.

Example 7B (Resin D2, Resin E2, and Resin C2 at Weight Ratio of 1/1/1)

(205) The fiber-reinforced resin molded body of Example 7B, which was prepared 90 days after preparation of the prepreg of Example 7B including the mixed resin of the resin D2, the resin E2, and the resin C2 (weight ratio: 1/1/1), had an appearance of “○”, a flexural strength of 880 MPa, a flexural modulus of 56 GPa, a flexural strength retention of 95%, and a flexural modulus retention of 95%. Thus, the deterioration in physical properties of the fiber-reinforced resin molded body due to storage of the prepreg was small.

(206) As described above, according to the present disclosure, it is possible to obtain a prepreg that is simple and inexpensive to prepare and has good storage stability. Further, a fiber-reinforced resin molded body having good quality prepared from the prepreg can be obtained.

(207) In Examples, only the flat plate shape is shown as the shape of the fiber-reinforced resin molded body, but in the present disclosure, the shape of the fiber-reinforced resin molded body (the shape of the mold) is not limited to the flat plate shape, and may be any shape such as a curved shape or an irregularity shape.

(208) In addition, the present aspect is not limited to the above Examples, and can be modified without departing from the spirit.

Third Aspect

(209) An embodiment of the third aspect will be described.

(210) 1. Fiber-Reinforced Molded Body **210**

(211) In the fiber-reinforced molded body **210**, a fiber substrate **211** is integrated by a thermosetting resin of a resin sheet **215** containing the thermosetting resin. The thermosetting resin has a viscosity of 2,000 Pa·Math.s or less at the curing reaction start temperature $T_b^{\circ}\text{C.}$, and a maximum viscosity of 1,000 Pa·Math.s or more in a temperature range of the curing reaction start temperature $T_b^{\circ}\text{C.}$ to 190°C.

(212) (1) Fiber Substrate **211**

(213) The fiber substrate **211** may be a single layer or a plurality of layers, and the number of layers is determined according to, for example, the use of the fiber-reinforced molded body **210**. In the

forms of FIGS. 14 and 15, the fiber substrate **211** composed of four layers is exemplified. Examples of the fiber substrate **211** include woven fabrics and nonwoven fabrics made of glass fibers, aramid fibers, basalt fibers, and carbon fibers, but are not particularly limited thereto. The fiber substrate **211** is preferably a carbon fiber woven fabric because it is excellent in lightweight and high rigidity. The carbon fiber woven fabric is preferably a woven fabric in which fibers are not unidirectional. Preferred examples thereof include plain weave, twill weave, satin weave, all of which are constituted of warp and weft, and triaxial weave which is constituted of threads in three directions. The carbon fiber woven fabric preferably has a fiber weight of 50 to 600 g/m.^{sup.2} from the viewpoint of impregnation of the thermosetting resin contained in the resin sheet **215** and rigidity of the fiber-reinforced molded body **210**.

(214) (2) Resin Sheet **215** Containing Thermosetting Resin

(215) As the thermosetting resin, a thermosetting resin that is solid at normal temperature (5° C. to 35° C.) is used in the production of the fiber-reinforced molded body **210**. The shape of the solid is not particularly limited. Examples of the solid shape include spherical, needle-shaped, and flake-shaped powders.

(216) The resin sheet **215** preferably includes a sheet substrate. When the resin sheet **215** includes the sheet substrate, the strength of the resin sheet **215** is increased, so that the handleability of the resin sheet **215** is improved. Even when the resin sheet **215** does not include the sheet substrate, the handleability is good as compared with the case of using a powder resin.

(217) The structure of the sheet substrate is not particularly limited. The sheet substrate preferably has a structure into which the melted resin can permeate. The structure into which the melted resin can permeate is not particularly limited, and examples thereof include a structure having communication pores. Preferably, the sheet substrate is not melted at the reaction start temperature (T_b) of the thermosetting resin.

(218) The sheet substrate is preferably one or more types selected from the group consisting of a foam, a nonwoven fabric, and a fiber sheet. When the sheet substrate has these structures, the thermosetting resin can be sufficiently retained in the space in the sheet substrate.

(219) The material of the sheet substrate is not particularly limited. The material of the sheet substrate is preferably one or more types selected from the group consisting of urethane, rayon, polyester, and carbon.

(220) Specifically, the sheet substrate is preferably one or more types selected from the group consisting of a urethane foam, a nonwoven fabric made of rayon and polyester (PET), a PET nonwoven fabric, and a carbon fiber sheet.

(221) The thickness of the resin sheet substrate is not particularly limited. The thickness of the resin sheet substrate is preferably 0.05 mm or more and 1.0 mm or less, and more preferably 0.08 mm or more and 0.7 mm or less, from the viewpoint of sufficiently retaining the thermosetting resin necessary for adhesion.

(222) The basis weight of the resin sheet substrate is not particularly limited. The basis weight of the resin sheet substrate is preferably 20 g/m.^{sup.2} or more and 50 g/m.^{sup.2} or less, and more preferably 30 g/m.^{sup.2} or more and 45 g/m.^{sup.2} or less.

(223) The resin sheet **215** is disposed in contact with the fiber substrate **211**. When the fiber substrate **211** is heat-compressed together with the resin sheet **215**, the thermosetting resin contained in the resin sheet **215** is melted, impregnated into the fiber substrate **211**, and cured. As a form of disposing the resin sheet **215** in contact with the fiber substrate **211**, the following forms are exemplified. When the fiber substrate **211** is a single layer, the resin sheet **215** is disposed on at least one of the upper surface and the lower surface of the single layer of the fiber substrate **211**. When the fiber substrate **211** has a plurality of layers, the resin sheet **215** is disposed on at least one surface, that is, at least one of the uppermost surface, the lowermost surface, and the lamination surface (between the fiber substrates) of the plurality of layers.

(224) The thermosetting resin preferably satisfies $30 \leq (T_b - T_a) \leq 100$ when the melting start

temperature is $T_a^\circ \text{C.}$ and the curing reaction start temperature is $T_b^\circ \text{C.}$ When the value of $(T_b - T_a)$ is in this range, the melted thermosetting resin can be sufficiently impregnated into the fiber substrate **211**, so that the fiber-reinforced molded body **210** having uniform physical properties can be obtained.

(225) The thermosetting resin has a minimum viscosity of 2,000 Pa.Math.s or less at a melting start temperature of $T_a^\circ \text{C.}$ or higher. The minimum viscosity is preferably 1,500 Pa.Math.s or less.

When the minimum viscosity is in this range, the melted thermosetting resin can be sufficiently impregnated into the fiber substrate **211**, so that the fiber-reinforced molded body **210** having uniform physical properties can be obtained. The lower limit value of the minimum viscosity is not particularly limited. The lower limit value of the minimum viscosity is preferably 0.005 Pa.Math.s.

(226) The minimum viscosity at the melting start temperature of $T_a^\circ \text{C.}$ or higher is the same as the viscosity at the curing reaction start temperature $T_b^\circ \text{C.}$

(227) The thermosetting resin preferably has a maximum viscosity of 1,000 Pa.Math.s or more in a temperature range of the curing reaction start temperature $T_b^\circ \text{C.}$ to 190°C. When the maximum viscosity is in this range, the melted thermosetting resin can be impregnated into the fiber substrate **211** and retained therein. As a result, the shapability of the fiber-reinforced molded body **210** is good, and sufficient strength can be obtained in a short time. The upper limit value of the maximum viscosity is not particularly limited, but the upper limit value is preferably 300,000 Pa.Math.s.

(228) The thermosetting resin preferably has a melting start temperature $T_a^\circ \text{C.}$ of 60 to 100°C. When the melting start temperature $T_a^\circ \text{C.}$ of the thermosetting resin is in this range, temperature control can be easily performed when the laminate in which the resin sheet **215** is disposed between at least any two adjacent fiber substrates **211** is heat-compressed to melt and cure the thermosetting resin.

(229) The thermosetting resin that can satisfy the melting start temperature $T_a^\circ \text{C.}$, the curing reaction start temperature $T_b^\circ \text{C.}$, the range of the value of $(T_b - T_a)$, the minimum viscosity, and the maximum viscosity as described above is preferably selected from the group consisting of a phenol resin, a mixed resin of a phenol resin and an epoxy resin, a mixed resin of a phenol resin and a cyanate resin, a mixed resin of an epoxy resin and a cyanate resin, and a mixed resin of a phenol resin, an epoxy resin, and a cyanate resin. The phenol resin is excellent in flame retardancy, and thus can impart excellent strength and flame retardancy to the fiber-reinforced molded body **210**.

(230) As the phenol resin, for example, a novolac-type powder phenol resin is preferably used. The physical properties of the phenol resin are not particularly limited. For example, a phenol resin having the following physical properties is suitably employed. Melting point: 80°C. or higher and 100°C. or lower

(231) As the epoxy resin, for example, a bisphenol A type solid resin is preferably used. The physical properties of the epoxy resin are not particularly limited. For example, an epoxy resin having the following physical properties is suitably employed. Epoxy equivalent: 400 g/eq or more and 1,000 g/eq or less Softening point: 60°C. or higher and 100°C. or lower Viscosity: 0.10 Pa.Math.s or more and 0.30 Pa.Math.s or less (25°C.)

(232) The cyanate resin is a thermosetting resin having a cyanato group, and is also called a cyanate monomer. The physical properties of the cyanate resin before curing are not particularly limited. For example, a cyanate resin having the following physical properties is suitably employed. Melting point: 75°C. or higher and 85°C. or lower Viscosity: 0.010 Pa.Math.s or more and 0.015 Pa.Math.s or less (80°C.)

(233) Various powder additives such as a pigment, an antibacterial agent, and an ultraviolet absorber may be added to the thermosetting resin as long as the viscosity and reactivity of the thermosetting resin are not affected.

(234) The basis weight of the thermosetting resin in the resin sheet **215** is not particularly limited. The basis weight of the thermosetting resin is preferably 200 g/m.sup.2 or more and 800 g/m.sup.2

or less, and more preferably 400 g/m.sup.2 or more and 600 g/m.sup.2 or less, from the viewpoint of securing the strength of the fiber-reinforced molded body **210** and from the viewpoint of not impairing the appearance.

(235) (3) Physical Properties of Fiber-Reinforced Molded Body **210**

(236) The flexural modulus (based on the method according to JIS K7074 A) of the fiber-reinforced molded body **210** is not particularly limited. The flexural modulus of the fiber-reinforced molded body **210** is preferably 40 GPa or more, more preferably 50 GPa or more from the viewpoint of high rigidity.

(237) The flexural strength (based on the method according to JIS K7074 A) of the fiber-reinforced molded body **210** is not particularly limited. The flexural strength of the fiber-reinforced molded body **210** is preferably 400 MPa or more, more preferably 800 MPa or more from the viewpoint of high strength.

(238) The specific gravity of the fiber-reinforced molded body **210** is not particularly limited. The specific gravity of the fiber-reinforced molded body **210** is preferably 1.10 or more and 1.80 or less, and more preferably 1.30 or more and 1.69 or less from the viewpoint of reducing the weight and not impairing the appearance.

(239) 2. Method for Producing Fiber-Reinforced Molded Body **210**

(240) The method for producing the fiber-reinforced molded body **210** of the present disclosure is performed by heat-compressing, by a mold, the fiber substrate **211** and the resin sheet **215** in a state of being stacked, so that a thermosetting resin is impregnated into the fiber substrate **211** and cured. With regard to the fiber substrate **211**, the resin sheet **215**, and the thermosetting resin, the description in the section “1. Fiber-reinforced molded body **210**” is referred to as it is.

(241) The form of disposing the resin sheet **215** is as described above, i.e., when the fiber substrate **211** is a single layer, the resin sheet is disposed on at least one of the upper surface and the lower surface of the fiber substrate, and when the fiber substrate **211** is a plurality of layers, the resin sheet is disposed on at least one of the uppermost surface, the lowermost surface, and the lamination surface (between the fiber substrates **211**) of the plurality of layers.

(242) When the resin sheet **215** is disposed on the lamination surface (between the fiber substrates **211**) of a plurality of layers of the fiber substrates **211**, the resin sheet **215** may be disposed on not only one lamination surface (between two adjacent fiber substrates **211**), but also on all lamination surfaces (between all fiber substrates) or on every predetermined number of lamination surfaces (between every predetermined number of fiber substrates **211**). The position of the surface where the sheet is to be disposed and the number of surfaces where the sheet is disposed are appropriately determined according to, for example, the number of laminated fiber substrates **211**.

(243) When the resin sheet **215** is disposed in contact with the upper surface or the lower surface of a single layer of the fiber substrate **211** or the uppermost surface or the lowermost surface of a plurality of layers of the fiber substrates **211**, a mold release sheet may be disposed between the resin sheet **215** and the mold surface of the mold for convenience of operation.

(244) An embodiment of a method for producing the fiber-reinforced molded body **210** including the fiber substrate **211** composed of four layers, shown in FIG. 14, will be described with reference to FIG. 15. In the following description of the production method, the plurality of fiber substrates **211** are denoted by a combined reference numeral of “**211**” and “alphabet”, such as “**211A**”, in order to easily grasp the vertical positional relationship in the plurality of fiber substrates **211**. Similarly, the plurality of resin sheets **215** are denoted by a combined reference numeral of “**215**” and “alphabet”, such as “**215A**”, in order to easily grasp the vertical positional relationship in the plurality of resin sheets **215**.

(245) In the embodiment shown in FIG. 15, when four fiber substrates **211A** to **211D** are laminated, resin sheets **215A** and **215B** are disposed between lower two fiber substrates **211A** and **211B** and upper two fiber substrates **211C** and **211D** (between the fiber substrate **211B** and the fiber substrate **211C**).

(246) The amount of the thermosetting resin contained in the resin sheets **215A** and **215B** is preferably adjusted so that the VF value (%) of the fiber-reinforced molded body **210** is 40 to 70%. The VF value (%) is a value calculated by (total weight of fiber substrate/density of fiber)/(volume of fiber-reinforced molded body)×100.

(247) A laminate of the fiber substrates **211A** to **211D**, the laminate being prepared by disposing the resin sheets **215A** and **215B** between the fiber substrate **211B** and the fiber substrate **211C** and laminating the fiber substrates, is sandwiched between a lower mold **231** and an upper mold **232** of a mold **230** that has been heated, and then the laminate is heat-compressed. The mold **230** is heated by a heating unit such as an electric heater to a temperature $T_c^{\circ}\text{C.}$ at which the thermosetting resin can be melted and cured.

(248) The temperature $T_c^{\circ}\text{C.}$ during heat compression (the temperature $T_c^{\circ}\text{C.}$ of the mold **30**) is preferably a temperature satisfying

$$[T_b+(T_b-T_a)/3]-15\leq T_c\leq[T_b+(T_b-T_a)/3]+20$$

(249) with respect to the melting start temperature $T_a^{\circ}\text{C.}$ and the curing reaction start temperature $T_b^{\circ}\text{C.}$ of the thermosetting resin. For example, when $T_a^{\circ}\text{C.}$ is 70°C. and $T_b^{\circ}\text{C.}$ is 130°C. , $T_c^{\circ}\text{C.}$ is 135°C. to 170°C.

(250) Pressurization (compression) of the fiber substrates **211A** to **211D** during heat compression by the mold **230** is preferably 2 MPa to 20 MPa so that the melted thermosetting resin contained in the resin sheets **215A** and **215B** between the fiber substrates **211** can be satisfactorily impregnated into the fiber substrates **211A** to **211D**.

(251) The compression percentage (%) of the fiber substrates **211A** to **211D** is a value calculated as (distance between mold surface of lower mold **231** and mold surface of upper mold **232**)/(total thickness of all layers of fiber substrates)×100, and is preferably 60 to 100%.

(252) The thermosetting resin contained in the resin sheets **215A** and **215B** between the fiber substrates **211** (between the fiber substrate **211B** and the fiber substrate **211C**) is melted through heating of the laminate by the mold **230**, and the melted thermosetting resin is impregnated into the fiber substrates **211B** and **211A** on the lower side and the fiber substrates **211C** and **211D** on the upper side through compression of the laminate. Then, the thermosetting resin that has been impregnated into the fiber substrates **211A** to **211D** is cured, whereby the fiber substrates **211A** to **211D** are integrated in a compressed state. Thus, the fiber-reinforced molded body **210** of FIG. **14**, which is shaped into the shape of the mold surface of the lower mold **231** and the upper mold **232**, is obtained.

(253) FIG. **16** shows an embodiment in which four fiber substrates **211A** to **211D** are laminated, resin sheets **215A** to **215C** are disposed between all the fiber substrates, and the laminate is heat-compressed by the mold **230**.

(254) The amount (total amount) of the thermosetting resin, the heating temperature of the mold **230**, the pressurization of the laminate, and the like are as described in the embodiment of FIG. **15**.

(255) In the embodiment shown in FIG. **17**, when ten fiber substrates **211A** to **211J** are laminated, five resin sheets **215A** to **215E** are disposed between lower five fiber substrates **211A** to **211E** and upper five fiber substrates **211F** to **211J** (between the fiber substrate **211E** and the fiber substrate **211F**).

(256) **3. Resin Sheet 215**

(257) The resin sheet **215** for producing the fiber-reinforced molded body **210** contains a thermosetting resin. That is, the resin sheet **215** supports the thermosetting resin in an uncured state. With regard to the resin sheet **215** and the thermosetting resin, the description in the section "1. Fiber-reinforced molded body **210**" is referred to as it is.

EXAMPLES

(258) Fiber-reinforced molded bodies of Examples 1C to 10C and Comparative Examples 1C and 2C were prepared as follows using the thermosetting resins shown in Tables 1 and 2. In Table 4, the characteristics of various sheet substrates used for preparing the fiber-reinforced molded body are

collectively described. The viscosity of the thermosetting resin was measured under the following conditions using a rheometer Rheosol-G3000, manufactured by UBM. 1) An amount of 0.4 g of a sample is formed into pellets (diameter ϕ : 18 mm, thickness: about 0.4 mm), and the formed pellets are sandwiched between parallel plates having a diameter ϕ of 18 mm. 2) The dynamic viscosity was measured at 2° C. intervals over a temperature range of 40° C. to 200° C. at a temperature rising rate of 5° C./min, a frequency of 1 Hz, and a rotation angle (strain) of 0.1 deg under constant temperature rising.

(259) TABLE-US-00001 TABLE 1 Melting Preparation Reaction start temperature start Type of Proportion of temperature of resin temperature thermosetting thermosetting (Ta) sheet (Tb) Tb – Ta resin resin [° C.] [° C.] [° C.] [° C.] Example Cyanate resin/ TA/AM-020-P/ 69 100 135 66 1C, 6C-10C Epoxy resin/ 50235D = 3/1/1 Phenol resin Example Cyanate resin/ TA/AM-030-P/ 95 100 135 40 2C Epoxy resin/ 50235D = 1/1/1 Phenol resin Example Phenol resin/ PR-50252/ 73 100 140 67 3C Epoxy resin 1001 = 1/1 Example Phenol resin/ PR-50235D/ 76 100 138 62 4C Cyanate resin TA = 1/1 Example Epoxy resin/ 1001/ 75 100 139 64 5C Cyanate resin TA = 1/1 Comparative Phenol resin PR-50699 72 80 91 19 Example 1C Comparative Phenol resin/ PR-50252/ 80 100 140 60 Example Phenol resin PR-50235D = 1/2 2C

(260) TABLE-US-00002 TABLE 2 Minimum Maximum viscosity viscosity Mold [Pa .Math. s] [Pa .Math. s] (Tb – Ta)/3 Tb + (Tb – Ta)/3 temperature (2,000 or less) (1,000 or more) [° C.] [° C.] [° C.] Example 59 8,768 22 157 160 1c, 6C-10C Example 1,500 209,004 13 148 150 2C Example 22 5,180 22 163 150 3C Example 475 51,895 21 159 170 4C Example 575 19,025 21 160 170 5C Comparative 118,908 164,468 6 97 100 Example 1C Comparative 21 260 20 160 160 Example 2C

(261) TABLE-US-00003 TABLE 3 Number of Sheet Number of laminated substrate resin fiber Flexural Flexural Molded of resin sheets substrates modulus strength Specific Thickness body Disposition sheet (sheet) (sheet) (GPa) (MPa) gravity (mm) appearance of resin Example PET 2 4 71.4 1,085 1.36 0.92 ○ Resin sheet is 1C nonwoven disposed between fabric fiber substrates at center Example PET 2 4 40.0 410 1.36 0.92 ○ Resin sheet is 2C nonwoven disposed between fabric fiber substrates at center Example PET 2 4 51.0 820 1.36 0.92 ○ Resin sheet is 3C nonwoven disposed between fabric fiber substrates at center Example PET 2 4 52.0 860 1.36 0.92 ○ Resin sheet is 4C nonwoven disposed between fabric fiber substrates at center Example PET 2 4 40.0 410 1.36 0.92 ○ Resin sheet is 5C nonwoven disposed between fabric fiber substrates at center Example PET 3 4 70.0 1,000 1.36 0.90 ○ Resin sheet is 6C nonwoven disposed between fabric respective fiber substrates Example PET 5 10 72.0 1,100 1.40 2.00 ○ Resin sheet is 7C nonwoven disposed between fabric fiber substrates at center Example Urethane 2 4 66.1 1,134 1.52 0.80 ○ Resin sheet is 8C resin disposed between foam fiber substrates at center Example Rayon/ 2 4 75.0 1,003 1.45 0.85 ○ Resin sheet is 9C Polyester disposed between nonwoven fiber substrates fabric at center Example Carbon 2 4 67.5 905 1.31 1.10 ○ Resin sheet is 10C fiber disposed between sheet fiber substrates at center Comp. PET 2 4 — — — — X Resin sheet is Example nonwoven disposed between 1C fabric fiber substrates at center Comp. PET 2 4 — — — — X Resin sheet is Example nonwoven disposed between 2C fabric fiber substrates at center

(262) TABLE-US-00004 TABLE 4 Weight Sheet substrate Thickness per unit area of resin sheet (mm) (g/m.sup.2) PET nonwoven fabric 0.08 45 Urethane resin foam 0.70 35 Rayon/Polyester 0.22 31 nonwoven fabric Carbon fiber sheet 0.34 31

1. Preparation of Fiber-Reinforced Molded Body

(1) Example 1C

(263) As the solid thermosetting resin, a mixed resin prepared by uniformly mixing a cyanate resin (manufactured by Mitsubishi Gas Chemical Company, Inc., product name: CYTESTER TA, average particle size: 100 μ m), an epoxy resin (manufactured by DIC Corporation, product name: AM-020-P, average particle size: 100 μ m), and a phenol resin (manufactured by Sumitomo Bakelite Co., Ltd., product name: PR-50235D, average particle size: 90 μ m) at a weight ratio of

3:1:1 was used.

(264) The characteristics of the mixed resin of Example 1C are as follows, and are described in Tables 1 and 2. The results of viscosity measurement of the mixed resin of Example 1C are shown in the graph of FIG. 18. Melting start temperature Ta: 69° C. Reaction start temperature Tb: 135° C. (Tb-Ta): 66° C. Minimum viscosity (minimum viscosity at melting start temperature of Ta° C. or higher): 59 Pa. Maximum viscosity (maximum viscosity in temperature range of curing reaction start temperature Tb° C. to 190° C.): 8,768 Pa. Value of (Tb-Ta)/3: 22° C. Value of Tb+(Tb-Ta)/3: 157° C.

(265) As the sheet substrate of the resin sheet, a sheet substrate obtained by cutting a PET nonwoven fabric (manufactured by Japan Vilene Company, Ltd., product name: JH-1004N1, weight per unit area: 45 g/m², thickness: 0.08 mm) shown in Table 4 into a size of 200 mm×250 mm was used.

(266) The solid thermosetting resin (20 g) was disposed on one sheet substrate to prepare a pre-molding sheet substrate.

(267) Next, one pre-molding sheet substrate was disposed on the molding surface of the lower mold of the mold heated to 100° C., and the mold was closed. Then, heat-compression was performed at a pressure of 1 MPa for 1 minute, thereby melting the solid thermosetting resin and causing the sheet substrate to support the thermosetting resin. Thereafter, cooling was performed to prepare a resin sheet.

(268) Two resin sheets thus prepared were prepared. The thickness of the resin sheet was adjusted by disposing a SUS spacer having a thickness of 1 mm between the lower mold and the upper mold, and disposing PET films having a thickness of 0.05 mm above and below the pre-molding sheet substrate.

(269) As the fiber substrate for reinforcement, four sheets of carbon fiber woven fabric (manufactured by Teijin Limited, product name: W-3101, weight per unit area: 200 g/m², thickness: 0.22 mm) each cut into 200 mm×250 mm were prepared. The weight of the cut carbon fiber woven fabric was 12 g per sheet. First, two sheets of carbon fiber woven fabric were disposed, two resin sheets were disposed thereon, and two sheets of carbon fiber woven fabric were further disposed thereon in this order to prepare a pre-molding laminate. In FIG. 15, a state of lamination is schematically illustrated. In Example 1C, as shown in FIG. 15, two resin sheets are disposed between fiber substrates (carbon fiber woven fabrics) at the center to form a pre-molding laminate.

(270) Next, the pre-molding laminate was disposed on the molding surface of the lower mold of the mold heated to 160° C., and the mold was closed. Then, heat-compression was performed at a pressure of 10 MPa for 10 minutes to melt and cure the solid thermosetting resin. When the solid thermosetting resin was melted and pressure was applied, the resin was impregnated into the fiber substrate of each layer, and then thermal curing of the solid thermosetting resin was completed. Thus, a fiber-reinforced molded body, in which the fiber substrate was integrated by the thermosetting resin of the resin sheet, was prepared. The thickness of the fiber-reinforced molded body was adjusted by disposing a SUS spacer having a thickness of 1 mm between the lower mold and the upper mold for press molding to adjust the distance between the lower mold and the upper mold.

(2) Example 2C

(271) A fiber-reinforced molded body was prepared in the same manner as in Example 1C except that a mixed resin prepared by uniformly mixing a cyanate resin (manufactured by Mitsubishi Gas Chemical Company, Inc., product name: CYTESTER TA, average particle size: 100 μm), an epoxy resin (manufactured by DIC Corporation, product name: AM-030-P, average particle size: 100 μm), and a phenol resin (manufactured by Sumitomo Bakelite Co., Ltd., product name: PR-50235D, average particle size: 90 μm) at a weight ratio of 1:1:1 was used as the solid thermosetting resin, and the mold temperature during molding of the fiber-reinforced molded body was 150° C.

(272) The characteristics of the mixed resin of Example 2C are as follows, and are described in

Tables 1 and 2. Melting start temperature Ta: 95° C. Reaction start temperature Tb: 135° C. (Tb-Ta): 40° C. Minimum viscosity (minimum viscosity at melting start temperature of Ta° C. or higher): 1,500 Pa.Math.s Maximum viscosity (maximum viscosity in temperature range of curing reaction start temperature Tb° C. to 190° C.) 209,004 Pa.Math.s Value of (Tb-Ta)/3: 13° C. Value of Tb+(Tb-Ta)/3: 148° C.

(3) Example 3C

(273) A fiber-reinforced molded body was prepared in the same manner as in Example 1C except that a mixed resin prepared by uniformly mixing a phenol resin (manufactured by Sumitomo Bakelite Co., Ltd., product name: PR-50252, average particle size: 30 μm) and an epoxy resin (manufactured by Mitsubishi Chemical Corporation, product name: jER-1001, average particle size: 100 μm) at a weight ratio of 1:1 was used as the solid thermosetting resin, and the mold temperature during molding of the fiber-reinforced molded body was 150° C.

(274) The characteristics of the mixed resin of Example 3C are as follows, and are described in Tables 1 and 2. Melting start temperature Ta: 73° C. Reaction start temperature Tb: 140° C.

(Tb-Ta): 67° C. Minimum viscosity (minimum viscosity at melting start temperature of Ta° C. or higher): 22 Pa.Math.s Maximum viscosity (maximum viscosity in temperature range of curing reaction start temperature Tb° C. to 190° C.) 5,180 Pa.Math.s Value of (Tb-Ta)/3: 22° C. Value of Tb+(Tb-Ta)/3: 163° C.

(4) Example 4C

(275) A fiber-reinforced molded body was prepared in the same manner as in Example 1C except that a mixed resin prepared by uniformly mixing a phenol resin (manufactured by Sumitomo Bakelite Co., Ltd., product name; PR-50235D, average particle size: 90 μm) and a cyanate resin (manufactured by Mitsubishi Gas Chemical Company, Inc., product name: CYTESTER TA, average particle size: 100 μm) at a weight ratio of 1:1 was used as the solid thermosetting resin, and the mold temperature during molding of the fiber-reinforced molded body was 170° C.

(276) The characteristics of the mixed resin of Example 4C are as follows, and are described in Tables 1 and 2. Melting start temperature Ta: 76° C. Reaction start temperature Tb: 138° C.

(Tb-Ta): 62° C. Minimum viscosity (minimum viscosity at melting start temperature of Ta° C. or higher): 475 Pa.Math.s Maximum viscosity (maximum viscosity in temperature range of curing reaction start temperature Tb° C. to 190° C.) 51,895 Pa.Math.s Value of (Tb-Ta)/3: 21° C. Value of Tb+(Tb-Ta)/3: 159° C.

(5) Example 5C

(277) A fiber-reinforced molded body was prepared in the same manner as in Example 1C except that a mixed resin prepared by uniformly mixing an epoxy resin (manufactured by Mitsubishi Chemical Corporation, product name: jER-1001, average particle size: 100 μm) and a cyanate resin (manufactured by Mitsubishi Gas Chemical Company, Inc., product name: CYTESTER TA, average particle size: 100 μm) at a weight ratio of 1:1 was used as the solid thermosetting resin, and the mold temperature during molding of the fiber-reinforced molded body was 170° C.

(278) The characteristics of the mixed resin of Example 5C are as follows, and are described in Tables 1 and 2. Melting start temperature Ta: 75° C. Reaction start temperature Tb: 139° C.

(Tb-Ta): 64° C. Minimum viscosity (minimum viscosity at melting start temperature of Ta° C. or higher): 575 Pa.Math.s Maximum viscosity (maximum viscosity in temperature range of curing reaction start temperature Tb° C. to 190° C.) 19,025 Pa.Math.s Value of (Tb-Ta)/3: 21° C. Value of Tb+(Tb-Ta)/3: 160° C.

(6) Example 6C

(279) A fiber-reinforced molded body was prepared in the same manner as in Example 1C except that same four fiber substrates for reinforcement as in Example 1C and the same three resin sheets as in Example 1C were prepared, and one resin sheet was disposed between respective layers of the fiber substrate as shown in FIG. 16.

(7) Example 7C

(280) A fiber-reinforced molded body was prepared in the same manner as in Example 1C except that a pre-molding substrate was prepared by preparing the same ten fiber substrates for reinforcement as in Example 1C, laminating five fiber substrates, disposing five resin sheets thereon, and further laminating the remaining five fiber substrates thereon. In FIG. 17, a state of lamination is schematically illustrated.

(8) Example 8C

(281) A fiber-reinforced molded body was prepared in the same manner as in Example 1C except that a urethane resin foam (manufactured by INOAC Corporation, product name: MF-50, weight per unit area: 35 g/m²) cut into a thickness of 0.7 mm and a planar size of 200 mm×300 mm was used as the sheet substrate.

(9) Example 9C

(282) A fiber-reinforced molded body was prepared in the same manner as in Example 1C except that a rayon/polyester nonwoven fabric (manufactured by Kuraray Trading Co., Ltd., product name: SF-30C, weight per unit area: 31 g/m²) cut into a thickness of 0.22 mm and a planar size of 200 mm×300 mm was used as the sheet substrate.

(10) Example 10C

(283) A fiber-reinforced molded body was prepared in the same manner as in Example 1C except that a carbon fiber sheet (manufactured by Awa Paper & Technological Company, Inc., product name: CARMIX C-2, weight per unit area: 31 g/m²) cut into a thickness of 0.34 mm and a planar size of 200 mm×300 mm was prepared as a sheet substrate.

(11) Comparative Example 1C

(284) A fiber-reinforced molded body was prepared in the same manner as in Example 1C except that a phenol resin (manufactured by Sumitomo Bakelite Co., Ltd., product name: PR-50699, average particle size: 30 μm) was used as the solid thermosetting resin, the mold temperature in the preparation of the resin sheet was 80° C., and the mold temperature during molding of the fiber-reinforced molded body was 100° C. The viscosity of the resin was high (the reaction was fast), the impregnation property of the resin was poor, and a uniform fiber-reinforced molded body could not be obtained.

(285) The characteristics of the resin of Comparative Example 1C are as follows, and are described in Tables 1 and 2. The results of viscosity measurement of the resin of Comparative Example 1C are shown in the graph of FIG. 18. Melting start temperature Ta: 72° C. Reaction start temperature Tb: 91° C. (Tb-Ta): 19° C. Minimum viscosity (minimum viscosity at melting start temperature of Ta° C. or higher): 118,908 Pa·s Maximum viscosity (maximum viscosity in temperature range of curing reaction start temperature Tb° C. to 190° C.) 164,468 Pa·s Value of (Tb-Ta)/3: 6° C. Value of Tb+(Tb-Ta)/3: 100° C.

(12) Comparative Example 2C

(286) A fiber-reinforced molded body was prepared in the same manner as in Example 1C except that a mixed resin of two types of phenol resins (weight ratio: 1:2) (product name: PR-50252, average particle size: 30 μm, manufactured by Sumitomo Bakelite Co., Ltd., and product name: PR-50235D, average particle size: 90 μm, manufactured by Sumitomo Bakelite Co., Ltd.) was used as the solid thermosetting resin, and the mold temperature during molding of the fiber-reinforced molded body was 160° C. The resin was not sufficiently cured, and deformation occurred at the time of demolding.

(287) The characteristics of the resin of Comparative Example 2C are as follows, and are described in Tables 1 and 2. The results of viscosity measurement of the resin of Comparative Example 2C are shown in the graph of FIG. 18. Melting start temperature Ta: 80° C. Reaction start temperature Tb: 140° C. (Tb-Ta): 60° C. Minimum viscosity (minimum viscosity at melting start temperature of Ta° C. or higher): 21 Pa·s Maximum viscosity (maximum viscosity in temperature range of curing reaction start temperature Tb° C. to 190° C.) 260 Pa·s Value of (Tb-Ta)/3: 20° C. Value of Tb+(Tb-Ta)/3: 160° C.

2. Physical Properties and the Like of Fiber-Reinforced Molded Body

(288) (1) Measurement Method

(289) The thickness (mm), flexural strength (MPa), and flexural modulus (GPa) of each of the fiber-reinforced molded bodies of Examples 1C to 10C and Comparative Examples 1C and 2C were measured, and the appearance thereof was evaluated. The results are shown in Table 3.

(290) A test piece was cut out from the fiber-reinforced molded body, and the flexural strength and the flexural modulus thereof were measured based on the method according to JIS K7074 A.

(291) The appearance was visually confirmed. In the evaluation of the appearance, presence of a defect such as deformation or non-uniformity of impregnation of resin on the surface of the fiber-reinforced molded body was visually confirmed, and the case where there was no defect was evaluated as “○”, and the case where there was a defect was evaluated as “×”

(292) The thickness of each part of the fiber-reinforced molded body was measured by observing the cross section of the fiber-reinforced molded body with a digital microscope VHX-5000 (manufactured by Keyence Corporation). The thickness in Table 3 is the thickness near the central part of the fiber-reinforced molded body.

(293) The specific gravity was calculated from the weight of the fiber-reinforced molded body and the volume of the fiber-reinforced molded body. The volume of the fiber-reinforced molded body was calculated from the thickness and area of the fiber-reinforced molded body.

(294) (2) Measurement Results

(295) The measurement results are shown in Table 3.

(296) The fiber-reinforced molded bodies of Examples 1C to 10C satisfy the following requirements (a) and (b). On the other hand, the fiber-reinforced molded body of Comparative Example 1C does not satisfy the requirement (a). In Comparative Example 1C which did not satisfy the requirement (a), the viscosity of the resin was high, and thus impregnation property of the resin was poor, and a uniform fiber-reinforced molded body could not be obtained. The fiber-reinforced molded body of Comparative Example 2C does not satisfy the requirement (b). In the fiber-reinforced molded body of Comparative Example 2C which did not satisfy the requirement (b), the curing of the resin was not sufficient, and deformation occurred at the time of demolding.

(297) In the fiber-reinforced molded bodies of Examples 1C to 10C which satisfied the requirements (a) and (b), a fiber-reinforced resin composite excellent in appearance, strength, and weight reduction could be obtained by a simple method without using a prepreg, by controlling the melting characteristics and curing characteristics of the solid thermosetting resin. In addition, it can be seen that in the fiber-reinforced molded bodies of Examples 1C to 10C, the resin sheet (resin-supporting sheet) can be prepared by a simple method, scattering of the powder can be prevented, an organic solvent or the like is not used in the production process, therefore, the working environment is excellent, and the problem of air pollution does not occur. Requirement (a): the viscosity at the curing reaction start temperature T_b ° C. (minimum viscosity) is 2,000 Pa·Math.s or less. Requirement (b): the maximum viscosity in a temperature range of the curing reaction start temperature T_b ° C. to 190° C. is 1,000 Pa·Math.s or more.

(298) In the fiber-reinforced molded bodies of Examples 1C to 10C which further satisfy the following requirement (c), the fiber substrate could be sufficiently impregnated with the melted thermosetting resin, and a fiber-reinforced molded body having uniform physical properties could be obtained. Requirement (c): $30 \leq (T_b - T_a) \leq 100$ is satisfied.

(299) The following inventions can also be grasped from the above Examples and Comparative Examples. The above description is appropriately incorporated for the description of specific matters of the following disclosure.

(300) A fiber-reinforced molded body obtained by impregnating a thermosetting resin into a laminate in which a fiber substrate and a sheet substrate different from the fiber substrate are laminated.

3. Effects of Examples

(301) According to the above Examples, a fiber-reinforced resin composite excellent in appearance, strength, and weight reduction could be obtained. In addition, it has been confirmed that a resin sheet can be prepared by a simple method, scattering of powder can be prevented, and an organic solvent or the like is not used in the production process, so that the working environment is excellent, and the problem of air pollution does not occur.

(302) The present aspect is not limited to Examples described in detail above, and various modifications or changes can be made.

Fourth Aspect

(303) An embodiment of the fourth aspect will be described. In a fiber-reinforced sandwich composite **310** of an embodiment shown in FIG. **19**, a fiber substrate **311** and a core component **315** are heat-compressed together with a thermosetting resin **321**, and bonded and integrated by curing of the thermosetting resin **321**.

(304) The fiber substrate **311** may be a single layer or a plurality of layers, and the number of layers is determined according to, for example, the use of the fiber-reinforced sandwich composite **310**. In the illustrated form, the fiber substrate **311** is composed of four layers. Examples of the fiber substrate **311** include woven fabrics and nonwoven fabrics made of glass fibers, aramid fibers, basalt fibers, and carbon fibers, but are not particularly limited thereto. The fiber substrate **311** is preferably a carbon fiber woven fabric because it is excellent in lightweight and high rigidity. The carbon fiber woven fabric is preferably a woven fabric in which fibers are not unidirectional. Preferred examples thereof include plain weave, twill weave, satin weave, all of which are constituted of warp and weft, and triaxial weave which is constituted of threads in three directions. The carbon fiber woven fabric preferably has a fiber weight of 50 to 600 g/m.² from the viewpoint of impregnation of the thermosetting resin **321** and rigidity of the fiber-reinforced sandwich composite **310**.

(305) The core component **315** has an effect of improving the strength of the fiber-reinforced sandwich composite **310**, and preferably has a 5% compressive stress value of 0.15 MPa or more. A more preferable range of the 5% compressive stress is 0.15 to 1 MPa. The 5% compressive stress is a stress when a core component cut into a size of 50×50 mm is compressed by 5% with respect to the thickness of the core component at a speed of 5 mm/min with a disk-shaped compression element having a diameter of 80 mm. An example of the stress measuring device is Autograph AG-X, manufactured by Shimadzu Corporation.

(306) The density (JIS 28807) of the core component **315** is preferably 20 to 120 kg/m.³.

(307) The core component **315** is preferably a foam having a closed cell structure. The foam having a closed cell structure has a structure in which cells (air bubbles) do not communicate with each other and are closed and independent from each other. When a foam having a closed cell structure is used as the core component **315**, the thermosetting resin melted in the production of the fiber-reinforced sandwich composite **310** does not easily permeate the core component **315** having a closed cell structure, and the surface appearance of the fiber-reinforced sandwich composite **310** becomes excellent. If the core component **315** is composed of a foam having an open-cell structure in which cells are opened and communicated with each other, the melted thermosetting resin easily permeates the core component **315** having an open-cell structure in the production of the fiber-reinforced sandwich composite **310**. As a result, the thermosetting resin may be reduced on the surface of the fiber-reinforced sandwich composite **310**, which may impair the appearance of the fiber-reinforced sandwich composite.

(308) Examples of the foam having a closed cell structure that constitutes the core component **315** include polyethylene terephthalate foam (PET foam), polymethacrylamide foam (PMI foam), polyvinyl chloride foam (PVC foam), and rigid polyurethane foam (rigid PU foam).

(309) The thickness of the core component **315** is determined according to the use of the fiber-reinforced sandwich composite **310** or the like, and is, for example, about 3 to 20 mm.

(310) The core component **315** is not limited to a single layer, and may be a multilayer. When the

core component **315** is a single layer and the fiber substrate **311** is a multilayer, the core component **315** is preferably located between the fiber substrates **311**.

(311) A method for producing the core component **315** is not particularly limited. Examples thereof include the following production methods.

(312) A method in which resin foam particles are charged into a mold, the resin foam particles are heated by a heat medium such as hot water or water vapor to be foamed, and the foamed particles are fused and integrated by foaming pressure of the resin foam particles, thereby producing a foam having a desired shape (in-mold foaming molding method).

(313) A method in which a resin is supplied to an extruder together with an air bubble adjusting agent and the like, melt-kneaded in the presence of a foaming agent such as a chemical foaming agent or a physical foaming agent, and the melt-kneaded product is extruded and foamed from the extruder, thereby producing a foam (extrusion foaming method).

(314) A method in which a massive foamable resin molded body containing a chemical foaming agent is produced, and the foamable resin molded body is foamed in a mold, thereby producing a foam.

(315) As the thermosetting resin **321**, a solid powdery thermosetting resin is used in the production of the fiber-reinforced sandwich composite **310**. The shape of the powder is not particularly limited, and examples thereof include a spherical shape, a needle shape, and a flake shape. The powder of the thermosetting resin **321** is preferably disposed in contact with at least the fiber substrate **311** and the core component **315**. For example, when the core component **315** is composed of a single layer, the powder of the thermosetting resin **321** may be disposed at least between the fiber substrate **311** and the core component **315**. When the fiber substrate **311** is composed of a plurality of layers, the powder of the thermosetting resin **321** may be disposed not only between the fiber substrate **311** and the core component **315** but also between the fiber substrates **311**. Further, when a plurality of layers of the core component **315** are laminated in contact with each other, the thermosetting resin powder **321** is preferably disposed between the fiber substrate **311** and the core component **315** and between the core components **315**.

(316) The powder of the thermosetting resin **321** is melted when heat-compressed together with the fiber substrate **311** and the core component **315**, and the melted thermosetting resin is impregnated into the fiber substrate **311** and is cured in contact with the core component **315**.

(317) The thermosetting resin **321** preferably has a viscosity of preferably 2,000 Pa·Math.s or less, and more preferably 1,500 Pa·Math.s or less at the curing reaction start temperature $T_b^{\circ}\text{C}$. When the viscosity at the curing reaction start temperature $T_b^{\circ}\text{C}$. is in this range, the melted thermosetting resin **321** can be sufficiently impregnated into the fiber substrate **311**. As a result, the fiber-reinforced sandwich composite **310** having uniform physical properties can be obtained.

(318) The thermosetting resin **321** preferably has a maximum viscosity of 1,000 Pa·Math.s or more in a temperature range of the curing reaction start temperature $T_b^{\circ}\text{C}$. to 190°C . When the maximum viscosity is in this range, the melted thermosetting resin **321** can be impregnated into the fiber substrate **311** and sufficiently cured. As a result, the shapability of the fiber-reinforced sandwich composite **310** is good, and sufficient strength can be obtained in a short time.

(319) The melting start temperature $T_a^{\circ}\text{C}$. and the curing reaction start temperature $T_b^{\circ}\text{C}$. of the thermosetting resin **321** are set to temperatures satisfying preferably $[T_b + (T_b - T_a)/3] - 15 \leq T_c \leq [T_b + (T_b - T_a)/3] + 20$, more preferably $[T_b + (T_b - T_a)/3] - 10 \leq T_c \leq [T_b + (T_b - T_a)/3] + 20$ with respect to the temperature $T_c^{\circ}\text{C}$. during heat compression. When the melting start temperature $T_a^{\circ}\text{C}$. and the curing reaction start temperature $T_b^{\circ}\text{C}$. of the thermosetting resin **321**, and the temperature $T_c^{\circ}\text{C}$. during heat compression are in this relationship, the powder of the thermosetting resin **321** is favorably melted during heat compression, and the thermosetting resin **321** is easily impregnated into the fiber substrate **311**. As a result, the fiber-reinforced sandwich composite **310** having uniform physical properties can be obtained.

(320) The value obtained by subtracting the melting start temperature $T_a^{\circ}\text{C.}$ from the curing reaction start temperature $T_b^{\circ}\text{C.}$ of the thermosetting resin **321** satisfies preferably $30\leq(T_b-T_a)\leq 100$, and more preferably $40\leq(T_b-T_a)\leq 70$. When the value of (T_b-T_a) is in this range, the melted thermosetting resin **321** can be sufficiently impregnated into the fiber substrate **311**, and the fiber-reinforced sandwich composite **310** having uniform physical properties can be obtained.

(321) The thermosetting resin **321** preferably has a melting start temperature $T_a^{\circ}\text{C.}$ of 60 to 100°C. When the melting start temperature $T_a^{\circ}\text{C.}$ of the thermosetting resin **321** is in this range, temperature control can be easily performed during heat compression.

(322) The thermosetting resin that can satisfy the melting start temperature $T_a^{\circ}\text{C.}$, the curing reaction start temperature $T_b^{\circ}\text{C.}$, the range of the value of (T_b-T_a) , the minimum viscosity, the maximum viscosity, and the like as described above is preferably selected from the group consisting of a phenol resin, a mixed resin of a phenol resin and an epoxy resin, a mixed resin of a phenol resin and a cyanate resin, and a mixed resin of a phenol resin, a cyanate resin, and an epoxy resin. The phenol resin is excellent in flame retardancy, and thus can impart excellent strength and flame retardancy to the fiber-reinforced sandwich composite **310**.

(323) Various powder additives such as a pigment, an antibacterial agent, and an ultraviolet absorber may be added to the thermosetting resin **321** as long as the viscosity and reactivity of the thermosetting resin are not affected.

(324) The production of the fiber-reinforced sandwich composite of the present disclosure can be performed by disposing the powder of the thermosetting resin **321** in contact with the fiber substrate **311** and the core component **315**; heat-compressing the fiber substrate **311** and the core component **315** together with the powder of the thermosetting resin **321** by a mold so that the powder of the thermosetting resin **321** is melted, impregnated into the fiber substrate **311**, and cured in a state being in contact with the core component **315**.

(325) An embodiment of a method for producing the fiber-reinforced sandwich composite **310** shown in FIG. **19** will be described with reference to FIG. **20**. In the following description of the production method, the fiber substrates **311** located at a plurality of positions are denoted by a combined reference numeral of number and alphabet, such as “**311A**”, in order to easily grasp the vertical positional relationship.

(326) In the embodiment shown in FIG. **20**, a pre-molding laminate is prepared by laminating two fiber substrates **311A** and **311B** among four fiber substrates **311A** to **311D**, disposing the thermosetting resin powder **321A** on the two fiber substrates, disposing the core component **315** thereon, disposing a thermosetting resin powder **321B** thereon, and laminating the remaining two fiber substrates **311C** and **311D** thereon.

(327) The particle sizes of the thermosetting resin powders **321A** and **321B** are preferably 10 to $500\text{ }\mu\text{m}$ from the viewpoint of ease of melting. The amounts of the thermosetting resin powders **321A** and **321B** are preferably adjusted so that the VF value (%) of the molded body excluding the foam portion of the core component is 40 to 70%. The VF value (%) is a value calculated by $(\text{total weight of fiber substrate}/\text{density of fiber})/(\text{volume of molded body excluding foam portion of core component})\times 100$.

(328) The prepared pre-molding laminate is sandwiched between a lower mold **331** and an upper mold **332** of a heated mold **330**, and is heat-compressed. A plastic mold release film or the like may be disposed on the mold surface of the mold. The mold **330** is heated by a heating unit such as an electric heater to $T_c^{\circ}\text{C.}$ that is the temperature during heat compression.

(329) Pressurization (compression) of the pre-molding laminate during heat compression by the mold **330** is preferably 2 to 20 MPa so that the melted thermosetting resin of the thermosetting resin powders **321A** and **321B** can be satisfactorily impregnated into the fiber substrates **311A** to **311D**.

(330) The thermosetting resin powders **321A** and **321B** located on both sides (upper and lower sides) of the core component **315** are melted through heating of the laminate by the mold **330**, and

the melted thermosetting resin are impregnated into the fiber substrates **311B** and **311A** on the lower side and the fiber substrates **311C** and **311D** on the upper side through compression of the pre-molding laminate. Then, the thermosetting resin impregnated into the fiber substrates **311A** to **311D** and in contact with the core component **315** is cured, whereby the fiber substrates **311A** to **311D** and the core component **315** between the fiber substrates **311B** and **311C** are bonded and integrated in a compressed state. Thus, the fiber-reinforced sandwich composite **310** of FIG. **19**, which is shaped into the shape of the mold surface of the lower mold **331** and the upper mold **332**, is obtained.

EXAMPLES

(331) Fiber-reinforced sandwich composites of Examples 1D to 7D and Comparative Examples 1D to 3D shown in FIG. **23** were prepared using the core components shown in FIG. **21** and the thermosetting resin powders shown in FIG. **22**.

(332) The 5% compressive stress of the core component was determined by measuring a stress when a core component cut into a size of 50×50 mm was compressed by 5% with respect to the thickness of the core component at a speed of 5 mm/min with a disk-shaped compression element having a diameter of 80 mm. The stress measuring device is Autograph AG-X, manufactured by Shimadzu Corporation.

(333) The viscosity of the thermosetting resin was measured under the following conditions using a rheometer Rheosol-G3000, manufactured by UBM. 1) An amount of 0.4 g of a sample is formed into pellets (diameter ϕ : 18 mm, thickness: about 0.4 mm), and the formed pellets are sandwiched between parallel plates having a diameter ϕ of 18 mm. 2) The dynamic viscosity was measured at 2° C. intervals over a temperature range of 40° C. to 200° C. at a temperature rising rate of 5° C./min, a frequency of 1 Hz, and a rotation angle (strain) of 0.1 deg under constant temperature rising.

(334) For the fiber-reinforced sandwich composites of Examples 1D to 7D and Comparative Examples 1D to 3D, the product appearance was evaluated, and the density, thickness, flexural strength, and flexural modulus were measured.

(335) For the product appearance, presence of a defect such as deformation or non-uniformity of impregnation of resin on the surface of the fiber-reinforced sandwich composite was visually confirmed, and the case where there was no defect was evaluated as “○”, and the case where there was a defect was evaluated as “×”

(336) The density was measured according to JIS 28807. The flexural strength and flexural modulus were measured based on the method according to JIS K7074 A.

Example 1D

(337) As the fiber substrate, four sheets of carbon fiber woven fabric (manufactured by Teijin Limited, product name: W-3101, weight per unit area: 200 g/m.sup.2, thickness: 0.22 mm) each cut into 210×297 mm were prepared. The weight of the cut fiber substrate was 12.5 g per sheet. Two cut fiber substrates were laminated, and 25 g of the following resin A3 as the thermosetting resin powder was disposed thereon substantially uniformly. A core component obtained by cutting a polyethylene terephthalate foam (3A, manufactured by Composites Airex, AIREX (T10)) having a closed cell structure into a size of 210×297 mm was disposed on the resin A3. Then, 25 g of the resin A3 was disposed thereon substantially uniformly, and the remaining two fiber substrates were laminated thereon. Thus, a pre-molding laminate was prepared.

(338) The resin A3 is a phenol resin (manufactured by Sumitomo Bakelite Co., Ltd., product name: PR-50252, average particle size: 30 μ m).

(339) The results of viscosity measurement of the resin A3 (Examples 1D, 6D, 7D) are shown in the graph of FIG. **24**.

(340) The pre-molding laminate was disposed on a molding surface (mold surface) of a lower mold of a mold heated to 150° C., and an upper mold of the mold was placed on the pre-molding laminate to close the mold. The laminate was heat-compressed at a pressure of 5 MPa for 10

minutes. The thermosetting resin powder is melted through heating, the melted thermosetting resin is impregnated into the fiber substrate of each layer through compression of the pre-molding laminate, and the thermosetting resin was completely cured in a state of being in contact with the core component. Thus, a fiber-reinforced sandwich composite of Example 1D, in which the fiber substrate and the core component are laminated and integrated by curing of the thermosetting resin, was prepared.

(341) The fiber-reinforced sandwich composite of Example 1D has a product appearance of “○”, a density of 0.24 g/cm³, a flexural strength of 23 MPa, and a flexural modulus of 6.2 GPa, and thus has a good appearance, high strength and rigidity, and light weight.

Example 2D

(342) A fiber-reinforced sandwich composite of Example 2D was prepared in the same manner as in Example 1D except that the following resin B3 was used as the thermosetting resin powder.

(343) The resin B3 is a phenol resin (manufactured by Sumitomo Bakelite Co., Ltd., product name: PR-310, average particle size: 30 μm).

(344) The fiber-reinforced sandwich composite of Example 2D has a product appearance of “○”, a density of 0.26 g/cm³, a flexural strength of 19 MPa, and a flexural modulus of 4.3 GPa, and thus has a good appearance, high strength and rigidity, and light weight.

Example 3D

(345) A fiber-reinforced sandwich composite of Example 3D was prepared in the same manner as in Example 1D except that a resin (25 g) prepared by uniformly mixing 12.5 g of the resin A3 and 12.5 g of the following resin C3 was used as the thermosetting resin powder, and the mold temperature was 170° C.

(346) As the resin C3, an epoxy resin (manufactured by Mitsubishi Chemical Corporation, product name: jER-1001) was pulverized in a mortar and used. The average particle size was 100 μm.

(347) The fiber-reinforced sandwich composite of Example 3D has a product appearance of “○”, a density of 0.23 g/cm³, a flexural strength of 24 MPa, and a flexural modulus of 6.8 GPa, and thus has a good appearance, high strength and rigidity, and light weight.

Example 4D

(348) A fiber-reinforced sandwich composite of Example 4D was prepared in the same manner as in Example 1D except that a resin (25 g) prepared by uniformly mixing 12.5 g of the following resin D3 and 12.5 g of the following resin E3 was used as the thermosetting resin powder, and the mold temperature was 160° C.

(349) As the resin D3, a phenol resin (manufactured by Sumitomo Bakelite Co., Ltd., product name: PR-50235D) was pulverized in a mortar and used. The average particle size was 90 μm.

(350) As the resin E3, a cyanate resin (manufactured by Mitsubishi Gas Chemical Company, Inc., product name: CYTESTER TA) was pulverized in a mortar and used. The average particle size was 100 μm.

(351) The fiber-reinforced sandwich composite of Example 4D has a product appearance of “○”, a density of 0.26 g/cm³, a flexural strength of 30 MPa, and a flexural modulus of 7.5 GPa, and thus has a good appearance, high strength and rigidity, and light weight.

Preparation of Example 5D

(352) A fiber-reinforced sandwich composite of Example 5D was prepared in the same manner as in Example 1D except that a resin (24.9 g) prepared by uniformly mixing 8.3 g of the resin D3, 8.3 g of the resin E3, and 8.3 g of the resin C3 was used as the thermosetting resin powder, and the mold temperature was 170° C.

(353) The fiber-reinforced sandwich composite of Example 5D has a product appearance of “○”, a density of 0.28 g/cm³, a flexural strength of 28 MPa, and a flexural modulus of 7.2 GPa, and thus has a good appearance, high strength and rigidity, and light weight.

Example 6D

(354) A fiber-reinforced sandwich composite of Example 6D was prepared in the same manner as

in Example 1D except that a polymethacrylamide foam (manufactured by Evonik Industries, product name: Rohacell (IG-31)) having a closed cell structure was used as the core component. (355) The fiber-reinforced sandwich composite of Example 6D has a product appearance of “○”, a density of 0.38 g/cm³, a flexural strength of 41 MPa, and a flexural modulus of 14.3 GPa, and thus has a good appearance, high strength and rigidity, and light weight.

Example 7D

(356) A fiber-reinforced sandwich composite of Example 7D was prepared in the same manner as in Example 1D except that a polyvinyl chloride foam (manufactured by GURIT, product name: CoreCell (HT-80)) having a closed cell structure was used as the core component.

(357) The fiber-reinforced sandwich composite of Example 7D has a product appearance of “○”, a density of 0.25 g/cm³, a flexural strength of 26 MPa, and a flexural modulus of 5.7 GPa, and thus has a good appearance, high strength and rigidity, and light weight.

Comparative Example 1D

(358) A fiber-reinforced sandwich composite of Comparative Example 1D was prepared in the same manner as in Example 1D except that a rigid polyurethane foam (manufactured by INOAC Corporation, product name: THERMAX (SII-25)) was used as the core component.

(359) The fiber-reinforced sandwich composite of Comparative Example 1D had a product appearance of “○”, a density of 0.21 g/cm³, a flexural strength of 5 MPa, and a flexural modulus of 0.6 GPa. The strength of the fiber-reinforced sandwich composite was insufficient because the 5% compressive strength of the core component used was low.

Comparative Example 2D

(360) A fiber-reinforced sandwich composite of Comparative Example 2D was prepared in the same manner as in Example 1D except that the following resin F3 was used as the thermosetting resin powder and the mold temperature was 100° C.

(361) The resin F3 is a phenol resin (manufactured by Sumitomo Bakelite Co., Ltd., product name: PR-50699, average particle size: 30 μm).

(362) The results of viscosity measurement of the resin F3 (Comparative Example 2D) are shown in the graph of FIG. 24.

(363) In Comparative Example 2D, the viscosity of the thermosetting resin was high (reaction was fast) and the impregnation property of the thermosetting resin into the fiber substrate was poor, and therefore, a uniform fiber-reinforced sandwich composite could not be obtained, and the density, thickness, flexural strength, and flexural modulus could not be measured.

Comparative Example 3D

(364) A fiber-reinforced sandwich composite of Comparative Example 3D was prepared in the same manner as in Example 1D except that a resin (24.9 g) prepared by uniformly mixing 8.3 g of the resin A3 and 16.6 g of the resin D3 was used as the thermosetting resin powder, and the mold temperature was 160° C.

(365) The results of viscosity measurement of the mixed resin of the resin A3 and the resin D3 (weight ratio: 1/2) (Comparative Example 3D) are shown in the graph of FIG. 24.

(366) In Comparative Example 3D, the thermosetting resin was not sufficiently cured, deformation occurred in the fiber-reinforced sandwich composite at the time of demolding, and the density, thickness, flexural strength, and flexural modulus could not be measured.

(367) As described above, according to the present aspect, there is no need for use of a prepreg, a solvent for a liquid thermosetting resin is not required, which eliminates a risk of environmental pollution, and there is no pot life of a thermosetting resin. It is therefore possible to obtain a fiber-reinforced sandwich composite that can be produced easily and inexpensively and safely without a risk of environmental pollution.

(368) The present aspect is not limited to Examples, and can be modified without departing from the spirit.

Fifth Aspect

(369) An embodiment of the fifth aspect will be described. A fiber-reinforced molded body **410** of the first embodiment shown in FIG. **25** is obtained by laminating and integrating a core material **411** and a fiber reinforcing material **415**, the core material **411** being obtained by impregnating a thermosetting resin into a resin foam and curing the thermosetting resin, and the fiber reinforcing material **415** being obtained by impregnating the thermosetting resin into a fiber substrate and curing the thermosetting resin. The fiber-reinforced molded body **410** is obtained by the method for producing a fiber-reinforced molded body of the present disclosure. The thermosetting resin impregnated into the resin foam and cured, and the thermosetting resin impregnated into the fiber substrate and cured are common thermosetting resins.

(370) The fiber reinforcing material **415** is preferably laminated on both surfaces of the core material **411** in order to prevent warpage of the fiber-reinforced molded body **410**. The number of laminated layers of the fiber reinforcing material **415** for each of both surfaces of the core material **411** is not limited to one, but may be two or more. In addition, the number of laminated fiber reinforcing materials **415** on the respective surfaces of the core material **411** may be different.

(371) As the resin foam, a resin foam having an open-cell structure is used. In the present disclosure, the resin foam having an open-cell structure means a resin foam mainly having an open-cell structure, and is not a resin foam mainly having a closed cell structure but slightly having an open-cell structure.

(372) The resin foam having an open-cell structure is not particularly limited, but a thermosetting resin foam having an open-cell structure is preferable. Examples thereof include urethane resin foams and melamine resin foams. When the fiber-reinforced molded body **410** is required to have flame retardancy, the resin foam having an open-cell structure is preferably a resin foam having flame retardancy, and the melamine resin foam is preferable.

(373) Examples of the fiber substrate include woven fabrics and nonwoven fabrics made of glass fibers, aramid fibers, basalt fibers, and carbon fibers, but are not particularly limited thereto. The fiber substrate is preferably a carbon fiber woven fabric because it is excellent in lightweight and high rigidity. The carbon fiber woven fabric is preferably a woven fabric in which fibers are not unidirectional. Preferred examples thereof include plain weave, twill weave, satin weave, all of which are constituted of warp and weft, and triaxial weave which is constituted of threads in three directions. The carbon fiber woven fabric preferably has a weight per unit area (fiber weight) of 50 to 600 g/m² from the viewpoint of impregnation of the thermosetting resin and rigidity of the fiber-reinforced molded body **410**.

(374) The thermosetting resin to be impregnated into the resin foam having an open-cell structure and the fiber substrate and cured has a viscosity of preferably 2,000 Pas or less, more preferably 1,500 Pa·s or less at the curing reaction start temperature T_b °C. When the viscosity at the curing reaction start temperature T_b °C. is in this range, the melted thermosetting resin **419** can be sufficiently impregnated into the resin foam having an open-cell structure and the fiber substrate in the production of the fiber-reinforced molded body **410**. As a result, the fiber-reinforced molded body **410** having uniform physical properties can be obtained. The thermosetting resin will be described in more detail in a method for producing the fiber-reinforced molded body **410** described later.

(375) The flexural modulus (based on the method according to JIS K7074 A) of the fiber-reinforced molded body **410** is preferably 24 GPa or more, more preferably 30 GPa or more.

(376) The flexural strength (based on the method according to JIS K7074 A) of the fiber-reinforced molded body **410** is preferably 300 MPa or more, more preferably 380 MPa or more.

(377) FIG. **26** relates to an embodiment of a method for producing the fiber-reinforced molded body **410**. In the following description of the production method, when members having the same names are disposed at a plurality of positions, the members are denoted by a combined reference numeral of number and alphabet, such as “**417A**”, in order to easily grasp the vertical positional relationship.

(378) In the embodiment shown in FIG. 26, a pre-molding laminate is prepared by disposing a resin foam **413** having an open-cell structure on one fiber substrate **417A**, disposing a powdery thermosetting resin **419** thereon, and further disposing a fiber substrate **417B** thereon.

(379) The position and the number of positions where the powdery thermosetting resin **419** is disposed are not limited, but it is preferable to dispose the thermosetting resin **419** at least at a position in contact with the resin foam **413**.

(380) The prepared pre-molding laminate is sandwiched between a lower mold **461** and an upper mold **462** of a heated mold **460**, and is heat-compressed. The mold **460** is heated by a heating unit such as an electric heater to $T_c^{\circ}\text{C}$. that is the temperature during heat compression.

(381) The powdery thermosetting resin **419** used in the method for producing the fiber-reinforced molded body **410** is the thermosetting resin as described above for the fiber-reinforced molded body **410**. The powdery thermosetting resin **419** has a viscosity of preferably 2,000 Pa·s or less, more preferably 1,500 Pa·s or less at the curing reaction start temperature $T_b^{\circ}\text{C}$. When the viscosity at the curing reaction start temperature $T_b^{\circ}\text{C}$. is in this range, the melted thermosetting resin **419** can be sufficiently impregnated into the resin foam **413** having an open-cell structure and the fiber substrates **417A** and **417B**, and a fiber-reinforced molded body **410** having uniform physical properties can be obtained.

(382) Further, the thermosetting resin **419** preferably has a maximum viscosity of 1,000 Pa·s or more in a temperature range of the curing reaction start temperature $T_b^{\circ}\text{C}$. to 190°C . When the maximum viscosity is in this range, the melted thermosetting resin **419** can be impregnated into the resin foam **413** having an open-cell structure and the fiber substrates **417A** and **417B** and sufficiently cured in the production of the fiber-reinforced molded body **410**. As a result, the shapability of the fiber-reinforced molded body **410** is good, and sufficient strength can be obtained in a short time.

(383) The melting start temperature $T_a^{\circ}\text{C}$. and the curing reaction start temperature $T_b^{\circ}\text{C}$. of the thermosetting resin **419** are set to temperatures satisfying preferably $[T_b + (T_b - T_a)/3] - 15 \leq T_c \leq [T_b + (T_b - T_a)/3] + 20$, more preferably $[T_b + (T_b - T_a)/3] - 10 \leq T_c \leq [T_b + (T_b - T_a)/3] + 20$ with respect to the temperature $T_c^{\circ}\text{C}$. during heat compression. When the melting start temperature $T_a^{\circ}\text{C}$. and the curing reaction start temperature $T_b^{\circ}\text{C}$. of the thermosetting resin, and the temperature $T_c^{\circ}\text{C}$. during heat compression are in this relationship, the thermosetting resin **419** is favorably melted during heat compression, and the thermosetting resin **419** is easily impregnated into the resin foam **413** having an open-cell structure and the fiber substrates **417A** and **417B**. As a result, the fiber-reinforced molded body **410** having uniform physical properties can be obtained.

(384) The value obtained by subtracting the melting start temperature $T_a^{\circ}\text{C}$. from the curing reaction start temperature $T_b^{\circ}\text{C}$. of the thermosetting resin **419** satisfies preferably $30 \leq (T_b - T_a) \leq 100$, and more preferably $40 \leq (T_b - T_a) \leq 70$. When the value of $(T_b - T_a)$ is in this range, the melted thermosetting resin **419** can be sufficiently impregnated into the resin foam **413** having an open-cell structure and the fiber substrates **417A** and **417B**, and a fiber-reinforced molded body **410** having uniform physical properties can be obtained.

(385) The thermosetting resin **419** preferably has a melting start temperature $T_a^{\circ}\text{C}$. of 60 to 100°C . When the melting start temperature $T_a^{\circ}\text{C}$. of the thermosetting resin **419** is in this range, temperature control can be easily performed during heat compression.

(386) The thermosetting resin that can satisfy the melting start temperature $T_a^{\circ}\text{C}$., the curing reaction start temperature $T_b^{\circ}\text{C}$., the range of the value of $(T_b - T_a)$, the viscosity at the curing reaction start temperature $T_b^{\circ}\text{C}$., the maximum viscosity, and the like as described above is preferably selected from the group consisting of a phenol resin, a mixed resin of a phenol resin and an epoxy resin, a mixed resin of a phenol resin and a cyanate resin, and a mixed resin of a phenol resin, a cyanate resin, and an epoxy resin. The phenol resin is excellent in flame retardancy, and thus can impart excellent strength and flame retardancy to the fiber-reinforced molded body.

(387) Various powder additives such as a pigment, an antibacterial agent, and an ultraviolet absorber may be added to the thermosetting resin **419** as long as the viscosity and reactivity of the thermosetting resin are not affected. The shape of powder of the thermosetting resin **419** is not particularly limited, and examples thereof include a spherical shape, a needle shape, and a flake shape. The particle size of the powdery thermosetting resin **419** is preferably 10 to 500 μm from the viewpoint of ease of melting.

(388) The amount of the thermosetting resin **419** is preferably such an amount that a resin proportion defined by Equation (A1) is 50 to 80%. When the resin proportion is in this range, the fiber-reinforced molded body **410** can have high strength.

[MathematicalFormula1]

$$(389) \quad \text{Resinproportion} = \frac{\text{Weightoffiber-reinforcedmoldedbody} - \text{Totalweightoffibersubstrateandresinfoambeforemolding}}{\text{Weightoffiber-reinforcedmoldedbody}} \times 100 \quad (\text{A1})$$

(390) Pressurization (compression) of the pre-molding laminate during heat compression by the mold **460** is preferably 2 to 20 MPa so that the melted thermosetting resin **419** can be satisfactorily impregnated into the resin foam **413** having an open-cell structure and the fiber substrates **417A** and **417B**. The compression of the pre-molding laminate is preferably set such that a compression percentage defined by Equation (A2) is 200 to 5,000%. When the compression percentage is in this range, the fiber-reinforced molded body **410** can have high strength.

[MathematicalFormula4]

$$(391) \quad \text{Compression percentage} = \frac{\text{Thicknessofresinfoambeforemolding} - \text{Thicknessofcorematerialoffiber-reinforcedmoldedbody}}{\text{Thicknessofcorematerialoffiber-reinforcedmoldedbody}} \times 100 \quad (\text{A2})$$

(392) At the time of heat-compressing the pre-molding laminate by the mold **460**, the thermosetting resin **419** is melted through heating of the pre-molding laminate by the mold **460**, and the melted thermosetting resin is impregnated into the resin foam **413** having an open-cell structure and the fiber substrates **417A** and **417B** through compression of the pre-molding laminate by the mold **460**. When the thermosetting resin **419** that has been impregnated into the resin foam **413** having an open-cell structure and the fiber substrates **417A** and **417B** is cured, the fiber substrates **417A** and **417B** and the resin foam **413** having an open-cell structure between the fiber substrates **417A** and **417B** are fixed in shape in a compressed state, and bonded and integrated. Thus, the fiber-reinforced molded body **410** of FIG. 25, which is shaped into the shape of the mold surface of the lower mold **461** and the upper mold **462**, is obtained.

(393) In the embodiment of the production method shown in FIG. 26, the powdery thermosetting resin **419** is disposed only on one side (one surface) of the resin foam **413** having an open-cell structure, but the fiber-reinforced molded body **410** may be produced by disposing the thermosetting resin on both sides of the resin foam **413**.

(394) FIG. 27 shows an embodiment in the case of producing the fiber-reinforced molded body **410** by disposing the powdery thermosetting resin **419** on both sides of the resin foam **413**.

(395) In the embodiment shown in FIG. 27, a pre-molding laminate is prepared by disposing a powdery thermosetting resin **419A** on one fiber substrate **419A**, disposing the resin foam **413** having an open-cell structure thereon, disposing a powdery thermosetting resin **419B** thereon, and further disposing the fiber substrate **417B** thereon.

(396) The prepared pre-molding laminate is sandwiched between the lower mold **461** and the upper mold **462** of the heated mold **460**, and heat-compressed to obtain the fiber-reinforced molded body **410** in FIG. 25 shaped into the shape of the mold surface of the lower mold **461** and the upper mold **462**. The resin foam **413** having an open-cell structure, the fiber substrates **417A** and **417B**, the powdery thermosetting resins **419A** and **419B**, the resin proportion, and production conditions of heating, compression, and the like are as described in the production embodiment of FIG. 26.

(397) A fiber-reinforced molded body **420** of the second embodiment shown in FIG. **28** is a case where two layers (two sheets) of the fiber reinforcing material **415** are laminated on both sides of the core material **411**. The core material **411** and the fiber reinforcing material **415** are as described in the fiber-reinforced molded body **410** of the first embodiment.

(398) In a fiber-reinforced molded body having two or more layers of the fiber reinforcing material **415**, the flexural modulus (based on the method according to JIS K7074 A) is also preferably 24 GPa or more, more preferably 30 GPa or more. The flexural strength (based on the method according to JIS K7074 A) of the fiber-reinforced molded body is preferably 300 MPa or more, more preferably 380 MPa or more.

(399) An embodiment shown in FIG. **29** will be described with respect to a method for producing the fiber-reinforced molded body **420** of the second embodiment.

(400) In the embodiment shown in FIG. **29**, a pre-molding laminate is prepared by laminating two fiber substrates **417A** and **417B**, disposing the resin foam **413** having an open-cell structure thereon, disposing the powdery thermosetting resin **419** thereon, and further laminating two fiber substrates **417C** and **417D** thereon.

(401) The prepared pre-molding laminate is sandwiched between the lower mold **461** and the upper mold **462** of the heated mold **460**, and heat-compressed to obtain the fiber-reinforced molded body **420** in FIG. **28** shaped into the shape of the mold surface of the lower mold **461** and the upper mold **462**. The mold **460** is heated by a heating unit such as an electric heater to $T_c^\circ \text{C.}$ that is the temperature during heat compression. The resin foam **413** having an open-cell structure, the fiber substrates **417A** to **417D**, the powdery thermosetting resin **419**, the resin proportion, and production conditions of heating, compression, and the like are as described in the production embodiment of FIG. **26**.

(402) A fiber-reinforced molded body **430** of the third embodiment shown in FIG. **30** will be described. The fiber-reinforced molded body **430** of the third embodiment is prepared by laminating and integrating the core material **411** and the fiber reinforcing material **415**, the core material **411** being obtained by impregnating a thermosetting resin into a resin foam and curing the thermosetting resin, and the fiber reinforcing material **415** being obtained by impregnating a thermosetting resin into a fiber substrate material and curing the thermosetting resin. The fiber-reinforced molded body **430** has a recess **431** and a protrusion **433** formed on the surface thereof. The fiber-reinforced molded body **430** of the third embodiment has the same configuration as that of the fiber-reinforced molded body **410** of the first embodiment except that the recess **431** and the protrusion **433** are formed.

(403) In a fiber-reinforced molded body having an irregularity shape on the surface thereof, the flexural modulus (based on the method according to JIS K7074 A) is also preferably 24 GPa or more, more preferably 30 GPa or more. The flexural strength (based on the method according to JIS K7074 A) of the fiber-reinforced molded body is preferably 300 MPa or more, more preferably 380 MPa or more.

(404) The irregularity shapes of the recess **431** and the protrusion **433** are not particularly limited as long as the compression percentage defined by Equation (A2) described above is in the range of 200 to 5,000%. For example, the depth, height, planar shape (linear, curved), and the like of the irregularity shape are appropriately determined according to use of the fiber-reinforced molded body **430** and the like. The irregularities may be formed not only on one surface but also on both surfaces of the fiber-reinforced molded body.

(405) An embodiment shown in FIG. **31** will be described with respect to a method for producing the fiber-reinforced molded body **430** of the third embodiment having irregularities formed on the surface thereof.

(406) In the production method of the embodiment shown in FIG. **31**, a pre-molding laminate is prepared by disposing the resin foam **413** having an open-cell structure on one fiber substrate **417A**, disposing the powdery thermosetting resin **419** thereon, and further disposing the fiber

substrate **417B** thereon.

(407) The prepared pre-molding laminate is sandwiched between a lower mold **471** of a heated mold **470** and an upper mold **472** having a recess forming protrusion **473** and a protrusion forming groove **475** formed in the mold surface thereof, and the laminate is heat-compressed. The recess forming protrusion **473** is a portion for forming the recess **431** of the fiber-reinforced molded body **430**, and the protrusion forming groove **475** is a portion for forming the protrusion **433** of the fiber-reinforced molded body **430**. The mold **470** is heated by a heating unit such as an electric heater to $T_c^\circ \text{C}$. that is the temperature during heat compression. The resin foam **413** having an open-cell structure, the fiber substrates **417A** and **417B**, the powdery thermosetting resin **419**, the resin proportion, and production conditions of heating, compression, and the like are as described in the production embodiment of FIG. 26.

(408) The pre-molding laminate is compressed into an irregularity shape conforming to the mold surface of the upper mold **472** by heat compression by the mold **470**, and the thermosetting resin **419** is melted and impregnated into the resin foam **413** having an open-cell structure and the fiber substrates **417A** and **417B**. Then, the impregnated thermosetting resin is cured whereby the fiber-reinforced molded body **430** in FIG. 30 shaped into the shape of the mold surface of the lower mold **471** and the upper mold **472**.

(409) In a fiber-reinforced molded body **440** of the fourth embodiment shown in FIG. 32, two layers (two sheets) of the fiber reinforcing material **415** are laminated on both sides of the core material **411**, and a recess **441** and a protrusion **443** are formed on the surface of the fiber-reinforced molded body **440**. The fiber-reinforced molded body **440** of the fourth embodiment has the same configuration as that of the fiber-reinforced molded body **420** of the second embodiment except that the recess **441** and the protrusion **443** are formed.

(410) In the fiber-reinforced molded body having irregularities on the surface thereof and having two or more layers of the fiber reinforcing material **415**, the flexural modulus (based on the method according to JIS K7074 A) is also preferably 24 GPa or more, more preferably 30 GPa or more. The flexural strength (based on the method according to JIS K7074 A) of the fiber-reinforced molded body is preferably 300 MPa or more, more preferably 380 MPa or more.

(411) An embodiment shown in FIG. 33 will be described with respect to a method for producing the fiber-reinforced molded body **440** of the fourth embodiment having irregularities formed on the surface thereof.

(412) In the production method of the embodiment shown in FIG. 33, a pre-molding laminate is prepared by laminating two fiber substrates **417A** and **417B**, disposing the resin foam **413** having an open-cell structure thereon, disposing the powdery thermosetting resin **419** thereon, and further laminating two fiber substrates **417C** and **417D** thereon.

(413) The prepared pre-molding laminate is sandwiched between the lower mold **471** of the heated mold **470** and the upper mold **472** having the recess forming protrusion **473** and the protrusion forming groove **475** formed in the mold surface thereof, and the laminate is heat-compressed. The recess forming protrusion **473** is a portion for forming the recess **441** of the fiber-reinforced molded body **440**, and the protrusion forming groove **475** is a portion for forming the protrusion **443** of the fiber-reinforced molded body **440**. The mold **470** is heated by a heating unit such as an electric heater to $T_c^\circ \text{C}$. that is the temperature during heat compression. The resin foam **413** having an open-cell structure, the fiber substrates **417A** to **417D**, the powdery thermosetting resin **419**, the resin proportion, production conditions of heating, compression, and the like are the same as those of the method for producing the fiber-reinforced molded body **420** of the second embodiment.

(414) The pre-molding laminate is compressed into an irregularity shape conforming to the mold surface of the upper mold **472** through heat compression by the mold **470**, and the thermosetting resin **419** is melted and impregnated into the resin foam **413** having an open-cell structure and the fiber substrates **417A** to **417D**. Then, the impregnated thermosetting resin is cured to obtain the

resin molded body **440** in FIG. **32** shaped into the shape of the mold surface of the lower mold **471** and the upper mold **472**.

(415) Also in the method for producing a fiber-reinforced molded body having an irregularity shape on the surface thereof, the powdery thermosetting resin may be disposed not only on one side of the resin foam having an open-cell structure but also on both sides of the resin foam. An embodiment of a production method in that case will be described with reference to FIG. **34**.

(416) The embodiment shown in FIG. **34** relates to the production of the fiber-reinforced molded body **440** of the fourth embodiment having an irregularity shape on the surface shown in FIG. **32**.

(417) In the embodiment of FIG. **34**, a pre-molding laminate is prepared by laminating two fiber substrates **417A** and **417B**, disposing the powdery thermosetting resin **419A** thereon, disposing the resin foam **413** having an open-cell structure thereon, disposing the powdery thermosetting resin **419B** thereon, and further laminating two fiber substrates **417C** and **417D** thereon.

(418) The prepared pre-molding laminate is sandwiched between the lower mold **471** of the heated mold **470** and the upper mold **472** having the recess forming protrusion **473** and the protrusion forming groove **475** formed in the mold surface thereof, and the laminate is heat-compressed. The resin foam **413** having an open-cell structure, the fiber substrates **417A** to **417D**, the powdery thermosetting resins **419A** and **419B**, the resin proportion, production conditions of heating, compression, and the like are as described above. The thermosetting resins **419A** and **419B** are melted and impregnated into the resin foam **413** having an open-cell structure and the fiber substrates **417A** to **417D**, through heat compression of the pre-molding laminate by the mold **470**. Then, the impregnated thermosetting resin is cured to obtain the fiber-reinforced molded body **440** in FIG. **32** shaped into the shape of the mold surface of the lower mold **471** and the upper mold **472**.

EXAMPLES

(419) Fiber-reinforced molded bodies of Examples and Comparative Examples shown in FIG. **36**, split across views labeled FIGS. **36A-36C**, were prepared using the powdery thermosetting resins shown in FIG. **35**.

(420) The viscosity of the thermosetting resin was measured under the following conditions using a rheometer Rheosol-G3000, manufactured by UBM. 1) An amount of 0.4 g of a sample was formed into pellets (diameter ϕ : 18 mm, thickness: about 0.4 mm), and the formed pellets were sandwiched between parallel plates having a diameter ϕ of 18 mm. 2) The dynamic viscosity was measured at 2° C. intervals over a temperature range of 40° C. to 200° C. at a temperature rising rate of 5° C./min, a frequency of 1 Hz, and a rotation angle (strain) of 0.1 deg under constant temperature rising. The melting start temperature (T_a) in FIG. **35** is a temperature indicating the maximum viscosity (the maximum viscosity in a temperature range of the curing reaction start temperature (T_b) or lower), the curing reaction start temperature (T_b) is a temperature indicating the minimum viscosity, and the maximum viscosity is a maximum viscosity in a temperature range of the curing reaction start temperature (T_b) to 190° C.

(421) For the fiber-reinforced molded bodies of Examples and Comparative Examples, the product appearance was evaluated, and the thickness (general portion, protrusion, recess) of the molded body, the thickness (general portion, protrusion, recess) of the core material, the compression percentage (general portion, protrusion, recess), the specific gravity, the weight, the resin proportion, the flexural strength, and the flexural modulus were calculated or measured.

(422) For the product appearance, presence of a defect such as deformation or non-uniformity of impregnation of resin on the surface of the fiber-reinforced molded body was visually checked, and the case where there was no defect was evaluated as “○”, the case where the appearance was slightly poor, such as the non-uniformity of the thickness was evaluated as “Δ”, and the case where there was a defect was evaluated as “×”.

(423) The compression percentage was calculated using Equation (A2) described above.

(424) The specific gravity was calculated from the weight of the fiber-reinforced molded body and

the volume of the fiber-reinforced molded body. The volume of the fiber-reinforced molded body was calculated from the thickness and area of the fiber-reinforced molded body.

(425) The resin proportion was calculated using Equation (A1) described above.

(426) The weights of the fiber substrate and the resin foam having an open-cell structure before molding were calculated by the following equations.

Weight of fiber substrate=weight per unit area×area of fiber-reinforced molded body×number of used fiber substrates

Weight of resin foam=density of resin foam×area of fiber-reinforced molded body×thickness of pre-molding resin foam

(427) The flexural strength and flexural modulus were measured based on the method according to JIS K7074 A.

Example 1E

(428) Examples 1E to 12E are examples of the fiber-reinforced molded body **410** shown in FIG. 25 and the production method of FIG. 26.

(429) In Example 1E, two sheets of carbon fiber woven fabric (manufactured by Teijin Limited, product name: W-3101, weight per unit area: 200 g/m², thickness: 0.22 mm) each cut into 200×300 mm were prepared as the fiber substrate. The weight of the cut fiber substrate was 12 g per sheet. In addition, as the resin foam having an open-cell structure, a melamine resin foam (manufactured by BASF SE, product name: Basotect G+, density: 9 kg/m³) cut into a thickness of 10 mm and a planar size of 200×300 mm (weight: 5.4 g) was prepared.

(430) A pre-molding laminate was prepared by laminating the melamine resin foam on one fiber substrate, disposing 65.0 g of the following resin A4 as the powdery thermosetting resin thereon substantially uniformly, and disposing another fiber substrate thereon.

(431) The resin A4 is a phenol resin (manufactured by Sumitomo Bakelite Co., Ltd., product name: PR-50252, average particle size: 30 μm).

(432) The results of viscosity measurements of the resin A4 (Examples 1E, 6E to 15E, 20E to 26E) are shown in the graph of FIG. 37.

(433) Next, the pre-molding laminate was disposed on a molding surface of a lower mold of a mold heated to 150° C., and the mold was closed. Then, heat compression was performed at a pressure of 10 MPa for 10 minutes. When the thermosetting resin was melted by heating and pressure was applied by the mold, the thermosetting resin was impregnated into the fiber substrate and the resin foam of each layer, and the curing reaction was completed. Thus, a fiber-reinforced molded body composed of an integrated laminate product including a fiber reinforcing material and a core material was prepared, in which the fiber reinforcing material was obtained by impregnating the thermosetting resin into the fiber substrate and curing the thermosetting resin, and the core material was obtained by impregnating the thermosetting resin into the resin foam having an open-cell structure and curing the thermosetting resin. The fiber-reinforced molded body was trimmed to 170×260 mm to obtain a fiber-reinforced molded body of Example 1.

(434) A SUS spacer having a thickness of 1 mm was disposed between the lower mold and the upper mold of the mold to set the distance between the lower mold and the upper mold to 1 mm, and the compressed thickness of the pre-molding laminate was adjusted to 1 mm.

(435) The numerical values and physical property values of the fiber-reinforced molded body of Example 1E are as shown in FIG. 36, split across views labeled FIGS. 36A-36C. As to the main numerical values and physical property values, the compression percentage is 1,686%, the specific gravity is 1.35, the resin proportion is 70%, the flexural strength is 500 MPa, the flexural modulus is 45 GPa, the product appearance is “○”. The fiber-reinforced molded body of Example 1E thus has a good appearance, high strength and rigidity, and light weight.

Example 2E

(436) A fiber-reinforced molded body of Example 2E was prepared in the same manner as in Example 1E except that the following resin B4 was used as the powdery thermosetting resin.

(437) The resin B4 is a phenol resin (manufactured by Sumitomo Bakelite Co., Ltd., product name: PR-310, average particle size: 30 μm).

(438) The fiber-reinforced molded body of Example 2E has a compression percentage of 1,686%, a specific gravity of 1.34, a resin proportion of 70%, a flexural strength of 400 MPa, a flexural modulus of 40 GPa, and a product appearance of “○”, and thus has a good appearance, high strength and rigidity, and light weight.

Example 3E

(439) A fiber-reinforced molded body of Example 3E was prepared in the same manner as in Example 1E except that a resin prepared by uniformly mixing the resin A4 and the following resin C4 at a weight ratio of 1:1 was used as the powdery thermosetting resin, and the mold temperature was 170° C.

(440) As the resin C4, an epoxy resin (manufactured by Mitsubishi Chemical Corporation, product name: jER-1001) was pulverized in a mortar and used. The average particle size was 100 μm .

(441) The fiber-reinforced molded body of Example 3E has a compression percentage of 1,686%, a specific gravity of 1.35, a resin proportion of 70%, a flexural strength of 820 MPa, a flexural modulus of 50 GPa, and a product appearance of “○”, and thus has a good appearance, high strength and rigidity, and light weight.

Example 4E

(442) A fiber-reinforced molded body of Example 4E was prepared in the same manner as in Example 1E except that a resin prepared by uniformly mixing the following resin D4 and the following resin E4 at a weight ratio of 1:1 was used as the powdery thermosetting resin, and the mold temperature was 160° C.

(443) As the resin D4, a phenol resin (manufactured by Sumitomo Bakelite Co., Ltd., product name: PR-50235D) was pulverized in a mortar and used. The average particle size was 90 μm .

(444) As the resin E4, a cyanate resin (manufactured by Mitsubishi Gas Chemical Company, Inc., product name: CYTESTER TA) was pulverized in a mortar and used. The average particle size was 100 μm .

(445) The fiber-reinforced molded body of Example 4E has a compression percentage of 1,686%, a specific gravity of 1.35, a resin proportion of 70%, a flexural strength of 850 MPa, a flexural modulus of 52 GPa, and a product appearance of “○”, and thus has a good appearance, high strength and rigidity, and light weight.

Example 5E

(446) A fiber-reinforced molded body of Example 5E was prepared in the same manner as in Example 1E except that a resin prepared by uniformly mixing the resin D4, the resin E4, and the resin C4 at a weight ratio of 1:1:1 was used as the powdery thermosetting resin, and the mold temperature was 170° C.

(447) The fiber-reinforced molded body of Example 5E has a compression percentage of 1,686%, a specific gravity of 1.35, a resin proportion of 70%, a flexural strength of 860 MPa, a flexural modulus of 53 GPa, and a product appearance of “○”, and thus has a good appearance, high strength and rigidity, and light weight.

Example 6E

(448) A fiber-reinforced molded body of Example 6E was prepared in the same manner as in Example 1E except that a melamine resin foam (manufactured by BASF SE, product name: Basotect G+, density: 9 kg/m³) cut into a thickness of 6 mm and a planar size of 200×300 mm (weight: 3.2 g) was used as the resin foam having an open-cell structure.

(449) The fiber-reinforced molded body of Example 6E has a compression percentage of 971%, a specific gravity of 1.35, a resin proportion of 70%, a flexural strength of 510 MPa, a flexural modulus of 46 GPa, and a product appearance of “○”, and thus has a good appearance, high strength and rigidity, and light weight.

Example 7E

(450) A fiber-reinforced molded body of Example 7E was prepared in the same manner as in Example 1E except that a melamine resin foam (manufactured by BASF SE, product name: Basotect G+, density: 9 kg/m³) cut into a thickness of 2 mm and a planar size of 200×300 mm (weight: 1.1 g) was used as the resin foam having an open-cell structure.

(451) The fiber-reinforced molded body of Example 7E has a compression percentage of 257%, a specific gravity of 1.35, a resin proportion of 70%, a flexural strength of 420 MPa, a flexural modulus of 40 GPa, and a product appearance of “○”, and thus has a good appearance, high strength and rigidity, and light weight.

Example 8E

(452) A fiber-reinforced molded body of Example 8E was prepared in the same manner as in Example 1E except that a melamine resin foam (manufactured by BASF SE, product name: Basotect G+, density: 9 kg/m³) cut into a thickness of 20 mm and a planar size of 200×300 mm (weight: 10.8 g) was used as the resin foam having an open-cell structure.

(453) The fiber-reinforced molded body of Example 8E has a compression percentage of 3,471%, a specific gravity of 1.36, a resin proportion of 71%, a flexural strength of 450 MPa, a flexural modulus of 42 GPa, and a product appearance of “○”, and thus has a good appearance, high strength and rigidity, and light weight.

Example 9E

(454) A fiber-reinforced molded body of Example 9E was prepared in the same manner as in Example 1E except that a melamine resin foam (manufactured by BASF SE, product name: Basotect G+, density: 9 kg/m³) cut into a thickness of 28 mm and a planar size of 200×300 mm (weight: 15.1 g) was used as the resin foam having an open-cell structure.

(455) The fiber-reinforced molded body of Example 9E has a compression percentage of 4,900%, a specific gravity of 1.36, a resin proportion of 71%, a flexural strength of 400 MPa, a flexural modulus of 39 GPa, and a product appearance of “○”, and thus has a good appearance, high strength and rigidity, and light weight.

Example 10E

(456) A fiber-reinforced molded body of Example 10E was prepared in the same manner as in Example 1E except that 48 g of the resin A was used as the powdery thermosetting resin.

(457) The fiber-reinforced molded body of Example 10E has a compression percentage of 2,074%, a specific gravity of 1.35, a resin proportion of 53%, a flexural strength of 410 MPa, a flexural modulus of 43 GPa, and a product appearance of “○”, and thus has a good appearance, high strength and rigidity, and light weight.

Example 11E

(458) A fiber-reinforced molded body of Example 11E was prepared in the same manner as in Example 1E except that 70 g of the resin A was used as the powdery thermosetting resin.

(459) The fiber-reinforced molded body of Example 11E has a compression percentage of 1,415%, a specific gravity of 1.37, a resin proportion of 76%, a flexural strength of 400 MPa, a flexural modulus of 42 GPa, and a product appearance of “○”, and thus has a good appearance, high strength and rigidity, and light weight.

Example 12E

(460) A fiber-reinforced molded body of Example 12E was prepared in the same manner as in Example 1E except that a urethane resin foam (manufactured by INOAC Corporation, product name: MF-50, density: 30 kg/m³) cut into a thickness of 10 mm and a planar size of 200×300 mm (weight: 18 g) was used as the resin foam having an open-cell structure.

(461) The fiber-reinforced molded body of Example 12E has a compression percentage of 1,686%, a specific gravity of 1.35, a resin proportion of 70%, a flexural strength of 450 MPa, a flexural modulus of 40 GPa, and a product appearance of “○”, and thus has a good appearance, high strength and rigidity, and light weight.

Example 13E

(462) Example 13E is a case where the powdery thermosetting resin is disposed on both sides of the resin foam having an open-cell structure, and is an example of the fiber-reinforced molded body **410** shown in FIG. 25 and the production method of FIG. 27.

(463) A fiber-reinforced molded body of Example 13E was prepared in the same manner as in Example 1E except that 32.5 g of the resin A4 as the powdery thermosetting resin was disposed above and below the melamine resin foam substantially uniformly.

(464) The fiber-reinforced molded body of Example 13E has a compression percentage of 1,686%, a specific gravity of 1.35, a resin proportion of 70%, a flexural strength of 500 MPa, a flexural modulus of 44 GPa, and a product appearance of “○”, and thus has a good appearance, high strength and rigidity, and light weight.

Example 14E

(465) Example 14E is a case where two fiber substrates are laminated on both sides of the resin foam having an open-cell structure, and is an example of the fiber-reinforced molded body **420** shown in FIG. 28 and the production method of FIG. 29.

(466) A fiber-reinforced molded body of Example 14E was prepared in the same manner as in Example 1E except that four sheets of carbon fiber woven fabric (manufactured by Teijin Limited, product name: W-3101, weight per unit area: 200 g/m.sup.2, thickness: 0.22 mm) each cut into 200×300 mm (the weight of the cut fiber substrate was 12 g per sheet) were prepared as the fiber substrate, a melamine resin foam was laminated on the two laminated fiber substrates, 57.0 g of the resin A4 as the powdery thermosetting resin was disposed thereon substantially uniformly, the remaining two fiber substrates were disposed thereon, and a SUS spacer having a thickness of 1.5 mm was disposed between the lower mold and the upper mold of the mold.

(467) The fiber-reinforced molded body of Example 14E has a compression percentage of 1,513%, a specific gravity of 1.40, a resin proportion of 62%, a flexural strength of 550 MPa, a flexural modulus of 48 GPa, and a product appearance of “○”, and thus has a good appearance, high strength and rigidity, and light weight.

Example 15E

(468) Examples 15E to 19E and Example 22E are examples of the fiber-reinforced molded body **430** (one fiber-reinforcing layer provided on each of both surfaces of the core material) having irregularities formed on the surface thereof as shown in FIG. 30 and the production method of FIG. 31.

(469) In Example 15E, two sheets of carbon fiber woven fabric (manufactured by Teijin Limited, product name: W-3101, weight per unit area: 200 g/m.sup.2, thickness: 0.22 mm) each cut into 200×300 mm were prepared as the fiber substrate. The weight of the cut fiber substrate was 12 g per sheet. In addition, as the resin foam having an open-cell structure, a melamine resin foam (manufactured by BASF SE, product name: Basotect G+, density: 9 kg/m.sup.3) cut into a thickness of 10 mm and a planar size of 200×300 mm (weight: 5.4 g) was prepared.

(470) A pre-molding laminate was prepared by laminating the melamine resin foam on one fiber substrate, disposing 65.0 g of the resin A4 as the powdery thermosetting resin thereon substantially uniformly, and disposing another fiber substrate thereon.

(471) Next, the pre-molding laminate was disposed on a molding surface of a lower mold of a mold heated to 150° C. Then, an upper mold is disposed on the pre-molding laminate, the mold was closed, and heat compression was performed at a pressure of 10 MPa for 10 minutes.

(472) On the mold surface of the upper mold of the mold, a recess forming protrusion (height 1 mm, width 10 mm) and a protrusion forming groove (depth 0.5 mm, width 10 mm) are formed from one end to the other end of the mold surface with a distance of 100 mm from each other.

(473) A SUS spacer having a thickness of 1 mm was disposed between the lower mold and the upper mold of the mold to set the distance between the lower mold and the upper mold to 1 mm, and the compressed thickness of the pre-molding laminate was adjusted to 1 mm.

(474) When the thermosetting resin was melted by heating during heat compression and pressure

was applied by the mold, the thermosetting resin was impregnated into the fiber substrate and the resin foam of each layer, and the curing reaction was completed. Thus, a fiber-reinforced molded body composed of an integrated laminate product including a fiber reinforcing material and a core material was prepared, in which the fiber reinforcing material was obtained by impregnating the thermosetting resin into the fiber substrate and curing the thermosetting resin, and the core material was obtained by impregnating the thermosetting resin into the resin foam having an open-cell structure and curing the thermosetting resin. One surface of the fiber-reinforced molded body has a recess shaped by the recess forming protrusion of the upper mold and a protrusion shaped by the protrusion forming groove. The fiber-reinforced molded body was trimmed to 170×260 mm to obtain a fiber-reinforced molded body of Example 15E.

(475) The numerical values and physical property values of the fiber-reinforced molded body of Example 15E are as shown in FIG. 36, split across views labeled FIGS. 36A-36C. As to the main numerical values and physical property values, the thickness is 1.0 mm and the compression percentage is 1,329% in the general portion without irregularities, the thickness is 2.0 mm and the compression percentage is 413% in the protrusion, the thickness is 0.6 mm and the compression percentage is 4,900% in the recess, the specific gravity is 1.35, the resin proportion is 70%, the flexural strength is 500 MPa, the flexural modulus is 46 GPa, and the product appearance is “o” in the entire molded body. The fiber-reinforced molded body of Example 15E thus has a good appearance, high strength and rigidity, and light weight. The flexural strength and the flexural modulus were measured for the general portion having no irregularities.

Example 16E

(476) A fiber-reinforced molded body of Example 16E was prepared in the same manner as in Example 15E except that the resin B4 was used as the powdery thermosetting resin.

(477) The fiber-reinforced molded body of Example 16E has a thickness of 1.0 mm and a compression percentage of 1,329% in the general portion without irregularities, a thickness of 2.0 mm and a compression percentage of 413% in the protrusion, a thickness of 0.6 mm and a compression percentage of 4,900% in the recess, a specific gravity of 1.34, a resin proportion of 70%, a flexural strength of 410 MPa, a flexural modulus of 40 GPa, and a product appearance of “o” in the entire molded body, and thus has a good appearance, high strength and rigidity, and light weight.

Example 17E

(478) A fiber-reinforced molded body of Example 17E was prepared in the same manner as in Example 15E except that a resin prepared by uniformly mixing the resin A4 and the resin C4 at a weight ratio of 1:1 was used as the powdery thermosetting resin, and the mold temperature was 170° C.

(479) The fiber-reinforced molded body of Example 17E has a thickness of 1.0 mm and a compression percentage of 1,329% in the general portion without irregularities, a thickness of 2.0 mm and a compression percentage of 413% in the protrusion, a thickness of 0.6 mm and a compression percentage of 4,900% in the recess, a specific gravity of 1.35, a resin proportion of 70%, a flexural strength of 820 MPa, a flexural modulus of 51 GPa, and a product appearance of “o” in the entire molded body, and thus has a good appearance, high strength and rigidity, and light weight.

Example 18E

(480) A fiber-reinforced molded body of Example 18E was prepared in the same manner as in Example 15E except that a resin prepared by uniformly mixing the resin D4 and the resin E4 at a weight ratio of 1:1 was used as the powdery thermosetting resin, and the mold temperature was 160° C.

(481) The fiber-reinforced molded body of Example 18E has a thickness of 1.0 mm and a compression percentage of 1,329% in the general portion without irregularities, a thickness of 2.0 mm and a compression percentage of 413% in the protrusion, a thickness of 0.6 mm and a

compression percentage of 4,900% in the recess, a specific gravity of 1.35, a resin proportion of 70%, a flexural strength of 860 MPa, a flexural modulus of 52 GPa, and a product appearance of “o” in the entire molded body, and thus has a good appearance, high strength and rigidity, and light weight.

Example 19E

(482) A fiber-reinforced molded body of Example 19E was prepared in the same manner as in Example 15E except that a resin prepared by uniformly mixing the resin D4, the resin E4, and the resin C4 at a weight ratio of 1:1:1 was used as the powdery thermosetting resin, and the mold temperature was 170° C.

(483) The fiber-reinforced molded body of Example 19E has a thickness of 1.0 mm and a compression percentage of 1,329% in the general portion without irregularities, a thickness of 2.0 mm and a compression percentage of 413% in the protrusion, a thickness of 0.6 mm and a compression percentage of 4,900% in the recess, a specific gravity of 1.35, a resin proportion of 70%, a flexural strength of 860 MPa, a flexural modulus of 53 GPa, and a product appearance of “o” in the entire molded body, and thus has a good appearance, high strength and rigidity, and light weight.

Example 20E

(484) Example 20E is an example of the fiber-reinforced molded body **440** (two fiber-reinforcing layers on each of both surfaces of the core material) having irregularities formed on the surface thereof shown in FIG. 32 and the production method of FIG. 33.

(485) A fiber-reinforced molded body of Example 20E was prepared in the same manner as in Example 15E except that four fiber substrates similar to those of Example 15E were prepared, the resin foam having an open-cell structure similar to that of Example 15E was laminated on two laminated fiber substrates, 57.0 g of the resin A4 as a powdery thermosetting resin was disposed thereon substantially uniformly, the remaining two fiber substrates were disposed thereon to prepare a pre-molding laminate, and a SUS spacer having a thickness of 1.5 mm was used.

(486) The fiber-reinforced molded body of Example 20E has a thickness of 1.5 mm and a compression percentage of 706% in the general portion without irregularities, a thickness of 2.5 mm and a compression percentage of 209% in the protrusion, a thickness of 1.1 mm and a compression percentage of 2,173% in the recess, a specific gravity of 1.40, a resin proportion of 62%, a flexural strength of 490 MPa, a flexural modulus of 42 GPa, and a product appearance of “o” in the entire molded body, and thus has a good appearance, high strength and rigidity, and light weight.

Example 21E

(487) Example 21E is an example of producing the fiber-reinforced molded body of FIG. 32 by disposing a powdery thermosetting resin on both sides of a resin foam having an open-cell structure as shown in FIG. 34.

(488) A fiber-reinforced molded body of Example 21E was prepared in the same manner as in Example 15E except that a pre-molding laminate was prepared by preparing four fiber substrates similar to those of Example 15E, disposing 29 g of the resin A4 as the powdery thermosetting resin on two laminated fiber substrates, laminating a resin foam having an open-cell structure similar to that of Example 15E thereon, disposing 29 g of the resin A4 as the powdery thermosetting resin thereon substantially uniformly, and disposing the remaining two fiber substrates thereon, and a SUS spacer having a thickness of 1.5 mm was used.

(489) The fiber-reinforced molded body of Example 21E has a thickness of 1.5 mm and a compression percentage of 706% in the general portion without irregularities, a thickness of 2.5 mm and a compression percentage of 209% in the protrusion, a thickness of 1.1 mm and a compression percentage of 2,173% in the recess, a specific gravity of 1.40, a resin proportion of 62%, a flexural strength of 490 MPa, a flexural modulus of 42 GPa, and a product appearance of “o” in the entire molded body, and thus has a good appearance, high strength and rigidity, and light weight.

weight.

Example 22E

(490) Example 22E is an example in which the type of the resin foam having an open-cell structure in Example 15E was changed.

(491) A fiber-reinforced molded body of Example 22E was prepared in the same manner as in Example 15E except that a urethane resin foam (manufactured by INOAC Corporation, product name: MF-50, density: 30 kg/m³) cut into a thickness of 10 mm and a planar size of 200×300 mm (weight: 18 g) was used as the resin foam having an open-cell structure.

(492) The fiber-reinforced molded body of Example 22E has a thickness of 1.0 mm and a compression percentage of 1,329% in the general portion without irregularities, a thickness of 2.0 mm and a compression percentage of 413% in the protrusion, a thickness of 0.6 mm and a compression percentage of 4,900% in the recess, a specific gravity of 1.35, a resin proportion of 70%, a flexural strength of 460 MPa, a flexural modulus of 38 GPa, and a product appearance of “o” in the entire molded body, and thus has a good appearance, high strength and rigidity, and light weight.

Example 23E

(493) A fiber-reinforced molded body of Example 23E was prepared in the same manner as in Example 1E except that the thickness of the resin foam having an open-cell structure was 30 mm.

(494) The fiber-reinforced molded body of Example 23E had a compression percentage of 5,257% and a product appearance of “A”.

Example 24E

(495) A fiber-reinforced molded body of Example 24E was prepared in the same manner as in Example 1E except that a urethane resin foam (manufactured by INOAC Corporation, product name: MF-50, density: 30 kg/m³) cut into a thickness of 1.5 mm and a planar size of 200×300 mm (weight: 18 g) was used as the thermosetting resin foam having an open-cell structure.

(496) The fiber-reinforced molded body of Example 24E had a compression percentage of 168%, a specific gravity of 1.35, a resin proportion of 70%, a flexural strength of 300 MPa, a flexural modulus of 24 GPa, and a product appearance of

Example 25E

(497) A fiber-reinforced molded body of Example 25E was prepared in the same manner as in Example 1E except that the amount of the powdery thermosetting resin used was adjusted so that the resin proportion was 45%.

(498) The fiber-reinforced molded body of Example 25E had a compression percentage of 2,074%, a specific gravity of 1.33, a resin proportion of 45%, a flexural strength of 350 MPa, a flexural modulus of 28 GPa, and a product appearance of “A”.

Example 26E

(499) A fiber-reinforced molded body of Example 26E was prepared in the same manner as in Example 1E except that the amount of the powdery thermosetting resin used was adjusted so that the resin proportion was 85%.

(500) The fiber-reinforced molded body of Example 26E had a compression percentage of 1,686%, a resin proportion of 85%, and a product appearance of “A”.

Comparative Example 1E

(501) Comparative Examples 1E and 2E are comparative examples with respect to Example 1E.

(502) In Comparative Example 1E, a fiber-reinforced molded body of Comparative Example 1E was prepared in the same manner as in Example 1E except that the following resin F4 was used as the powdery thermosetting resin, and the mold temperature was 100° C.

(503) The resin F4 is a phenol resin (manufactured by Sumitomo Bakelite Co., Ltd., product name: PR-50699, average particle size: 30 μm).

(504) The results of viscosity measurements of the resin F4 (Comparative Examples 1E, 3E) are shown in the graph of FIG. 37.

(505) The compression percentage of the fiber-reinforced molded body of Comparative Example 1E was 1,686%, but the minimum viscosity of the thermosetting resin was high (reaction was fast), and the thermosetting resin was not easily impregnated into the fiber substrate. As a result, the fiber-reinforced molded body could not be obtained, and physical property values and the like could not be measured.

Comparative Example 2E

(506) A fiber-reinforced molded body of Comparative Example 2E was prepared in the same manner as in Example 1E except that a resin prepared by uniformly mixing the resin A4 and the resin D4 at a weight ratio of 1:2 was used as the powdery thermosetting resin, and the mold temperature was 160° C.

(507) The results of viscosity measurement of the mixed resin of the resin A4 and the resin D4 (weight ratio: 1/2) (Comparative Examples 2E, 4E) are shown in the graph of FIG. 37.

(508) The compression percentage of the fiber-reinforced molded body of Comparative Example 2E was 1,686%, but the maximum viscosity of the thermosetting resin in the temperature range of Tb to 190° C. was low, the thermosetting resin was not cured, and deformation occurred in the fiber-reinforced molded body at the time of demolding. Therefore, physical property values and the like could not be measured.

Comparative Example 3E

(509) Comparative Examples 3E and 4E are comparative examples with respect to Example 15E.

(510) In Comparative Example 3E, a fiber-reinforced molded body of Comparative Example 3E was prepared in the same manner as in Example 15E except that the resin F4 was used as the powdery thermosetting resin, and the mold temperature was 100° C.

(511) In the fiber-reinforced molded body of Comparative Example 3E, the minimum viscosity of the thermosetting resin was high (reaction was fast), and the thermosetting resin was not easily impregnated into the fiber substrate and the resin foam having an open-cell structure. As a result, a fiber-reinforced molded body could not be obtained, and physical property values and the like could not be measured.

Comparative Example 4E

(512) A fiber-reinforced molded body of Comparative Example 4E was prepared in the same manner as in Example 15E except that a resin prepared by uniformly mixing the resin A4 and the resin D4 at a weight ratio of 1:2 was used as the powdery thermosetting resin, and the mold temperature was 160° C.

(513) In the fiber-reinforced molded body of Comparative Example 4E, the maximum viscosity of the thermosetting resin in the temperature range of Tb to 190° C. was low, and the thermosetting resin was not cured. As a result, deformation occurred in the fiber-reinforced molded body at the time of demolding, and physical property values and the like could not be measured.

(514) As described above, according to the present aspect, a fiber-reinforced molded body can be obtained without using a prepreg. Further, according to the present aspect, a fiber-reinforced molded body shaped into an irregularity shape can also be obtained.

(515) The present aspect is not limited to Examples, and can be modified without departing from the spirit.

REFERENCE SIGNS LIST

Description of Reference Numerals in Description of First Aspect

(516) **10** Fiber-reinforced resin molded body **11, 11A to 11F** Fiber substrate **15** Thermosetting resin **15A to 15E** Thermosetting resin powder **30** Mold **31** Lower mold **32** Upper mold

Description of Reference Numerals in Description of Second Aspect

(517) **110, 120, 130** Prepreg **111, 111A to 111D** Fiber substrate **115** Thermosetting resin **115A to 115C** Thermosetting resin powder **141A, 141B** Plastic mold release film **150** Mold **151** Lower mold **152** Upper mold

Description of Reference Numerals in Description of Third Aspect

(518) **210** Fiber-reinforced molded body **211** Fiber substrate **215** Resin sheet **230** Mold **231** Lower mold **232** Upper mold
 Description of Reference Numerals in Description of Fourth Aspect
 (519) **310** Fiber-reinforced sandwich composite **311**, **311A** to **311D** Fiber substrate **315** Core component **321** Thermosetting resin **321A**, **321B** Thermosetting resin powder **330** Mold **331** Lower mold **332** Upper mold
 Description of Reference Numerals in Description of Fifth Aspect
 (520) **410**, **420**, **430**, **440** Fiber-reinforced molded body **411** Core material **413** Resin foam having open-cell structure **415** Fiber reinforcing material **417**, **417A** to **417D** Fiber substrate **419**, **419A**, **419B** Powdery thermosetting resin **431**, **441** Recess **433**, **443** Protrusion **460**, **470** Mold **461**, **471** Lower mold **462**, **472** Upper mold **473** Recess forming protrusion **475** Protrusion forming groove

Claims

1. A fiber-reinforced resin molded body obtained by heat-compressing a fiber substrate together with a thermosetting resin so that the thermosetting resin is impregnated into the fiber substrate and cured, wherein the thermosetting resin cured is obtained by melting a powder of the thermosetting resin disposed in contact with the fiber substrate and curing the thermosetting resin during the heat compression, and any one of the following (1) to (5) is satisfied: (1) the fiber-reinforced resin molded body has a flexural strength (JIS K 7074 A method) of 450 MPa or more; (2) the fiber substrate includes a basalt fiber; (3) the thermosetting resin includes a cyanate resin; (4) a value of $(T_b - T_a)$ of the thermosetting resin satisfies $30 \leq (T_b - T_a) \leq 100$, wherein a melting start temperature of the thermosetting resin is $T_a^\circ \text{C.}$ and a curing reaction start temperature of the thermosetting resin is $T_b^\circ \text{C.}$; and (5) the melting start temperature $T_a^\circ \text{C.}$ of the thermosetting resin is 60 to 100°C.
2. A vehicle or an airframe comprising a fiber-reinforced resin molded body obtained by heat-compressing a fiber substrate together with a thermosetting resin so that the thermosetting resin is impregnated into the fiber substrate and cured, wherein the thermosetting resin cured is obtained by melting a powder of the thermosetting resin disposed in contact with the fiber substrate and curing the thermosetting resin during the heat compression, and any one of the following (1) to (5) is satisfied: (1) the fiber-reinforced resin molded body has a flexural strength (JIS K 7074 A method) of 450 MPa or more; (2) the fiber substrate includes a basalt fiber; (3) the thermosetting resin includes a cyanate resin; (4) a value of $(T_b - T_a)$ of the thermosetting resin satisfies $30 \leq (T_b - T_a) \leq 100$, wherein a melting start temperature of the thermosetting resin is $T_a^\circ \text{C.}$ and a curing reaction start temperature of the thermosetting resin is $T_b^\circ \text{C.}$; and (5) the melting start temperature $T_a^\circ \text{C.}$ of the thermosetting resin is 60 to 100°C.
3. A method for producing the fiber-reinforced resin molded body according to claim 1 by heat-compressing a fiber substrate together with a thermosetting resin so that the thermosetting resin is impregnated into the fiber substrate and cured, the method including: disposing a powder of the thermosetting resin in contact with the fiber substrate; and heat-compressing the fiber substrate together with the powder of the thermosetting resin by a mold so that the powder of the thermosetting resin is melted, impregnated into the fiber substrate, and cured.
4. A method for producing the fiber-reinforced resin molded body according to claim 1 by heat-compressing a fiber substrate together with a thermosetting resin so that the thermosetting resin is impregnated into the fiber substrate and cured, the method comprising: disposing a powder of the thermosetting resin in contact with the fiber substrate; and heat-compressing the fiber substrate together with the powder of the thermosetting resin by a mold so that the powder of the thermosetting resin is melted, impregnated into the fiber substrate, and cured, wherein a temperature $T_c^\circ \text{C.}$ of the mold is a temperature satisfying $[T_b + (T_b - T_a)/3] - 15 \leq T_c \leq [T_b + (T_b - T_a)/3] + 20$ wherein a melting start temperature of the

thermosetting resin is T_a ° C. and a curing reaction start temperature of the thermosetting resin is T_b ° C.
